ImPRESS Statistical Analysis Report

Baseline Review for the Newly Diagnosed Glioblastoma Cohort

ImPRESS Statistical Team

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Contents

Introduction

The ImPRESS study is a prospective investigation of losartan as an adjunctive therapy for patients with glioblastoma. According to Protocol Version 7.0 (17-Dec-2024), the trial collects comprehensive demographic, clinical, imaging, and quality-of-life information prior to treatment initiation. The statistical objectives for the interim analyses focus on characterising baseline status, neurological function, and tumour burden before randomisation to evaluate comparability across planned treatment arms.

This report summarises the currently available baseline data with emphasis on clinical status, imaging assessments, and prior treatments that may influence subsequent outcomes. The material is intended to provide early insight into data quality and cohort composition ahead of formal efficacy or safety analyses.

Randomisation Status

The present dataset is built on *sham randomisation* generated from the most recent export (see `_targets.R` workflow). No true treatment assignments have been unlocked. All findings in this document should therefore be interpreted as preliminary and descriptive only. Final efficacy comparisons will not be conducted until the database is locked and genuine randomisation codes are released.

Cohort Scope

The results herein are restricted to participants enrolled in the **newly diagnosed glioblastoma (AN) cohort**. Subjects with recurrent disease are excluded from all summaries, figures, and tables in this draft. A combined-cohort analysis will be delivered once recruitment and cleaning are complete for both strata.

Future revisions of this report will:

- incorporate the validated randomisation list once database lock is confirmed,
- integrate data from the recurrent glioblastoma cohort,
- extend the analyses to longitudinal efficacy and safety endpoints.

Baseline Characteristics

This section summarises baseline characteristics for the newly diagnosed glioblastoma cohort using the `adbl` ADaM dataset.

Demographics and Baseline characteristics

	all obs
Age (years)	
n	54
Mean (SD)	52.6 (10.2)
Median	55.0
Min - Max	26.0 - 69.0
Sex	
n	54
Female	29 (53.7%)
Male	25 (46.3%)
Cohort	
n	54
Newly diagnosed glioblastoma	54 (100%)
Country	
n	54
NOR	54 (100%)
Childbearing potential?	
n	29
No	13 (44.8%)
Yes	16 (55.2%)
Diagnosis	
n	54

Demographics and Baseline characteristics

	all obs
Newly diagnosed glioblastoma	54 (100%)
Time from surgery to randomisation (days)	
n	54
Mean (SD)	27.4 (7.3)
Median	25.5
Min - Max	13.0 - 48.0
IDH1/2 mutation	
n	54
Negative	47 (87%)
Positive	7 (13%)
MGMT status	
n	54
Negative	30 (55.6%)
Positive	24 (44.4%)
Pregnancy test result	
n	13
Negative	13 (100%)
ECOG Score	
n	54
Mean (SD)	0.3 (0.5)
Median	0.0
Min - Max	0.0 - 2.0
ECOG Score (categorical)	
n	54
ECOG 0	40 (74.1%)
ECOG 1	13 (24.1%)
ECOG 2	1 (1.9%)
Karnofsky Performance Scale	
n	54
Mean (SD)	95.0 (8.0)
Median	100.0
Min - Max	60.0 - 100.0
Karnofsky Performance Scale (categorical)	
n	54

Demographics and Baseline characteristics

	all obs
KPS 100	34 (63%)
KPS 60	1 (1.9%)
KPS 80	4 (7.4%)
KPS 90	15 (27.8%)
Total NANO score	
n	54
Mean (SD)	1.0 (1.3)
Median	0.0
Min - Max	0.0 - 4.0
EORTC QLQ-C30 score	
n	53
Mean (SD)	56.3 (8.6)
Median	54.0
Min - Max	41.0 - 77.0
EORTC QLQ-BN20 score	
n	53
Mean (SD)	30.8 (6.6)
Median	30.0
Min - Max	20.0 - 49.0
Systolic blood pressure	
n	54
Mean (SD)	130.0 (15.4)
Median	128.5
Min - Max	106.0 - 180.0
Diastolic blood pressure	
n	54
Mean (SD)	81.9 (7.1)
Median	82.0
Min - Max	67.0 - 97.0
Pulse Rate	
n	54
Mean (SD)	80.7 (15.7)
Median	78.0
Min - Max	54.0 - 114.0

Demographics and Baseline characteristics

	all obs
Weight (kg)	
n	54
Mean (SD)	78.8 (13.2)
Median	78.3
Min - Max	48.2 - 116.0
BMI	
n	51
Mean (SD)	25.0 (4.2)
Median	23.9
Min - Max	18.6 - 39.9
Height (cm)	
n	51
Mean (SD)	176.5 (8.8)
Median	175.0
Min - Max	158.0 - 194.0
Has the patient received chemotherapy	
n	54
No	54 (100%)
Has the patient received immunotherapy	
n	54
No	54 (100%)
Has the patient received radiation therapy to the brain?	
n	54
No	54 (100%)
CNS surgery	
n	54
No	3 (5.6%)
Yes	51 (94.4%)
Steroids at time of MRI	
n	54
No	42 (77.8%)
Yes	12 (22.2%)
Mean relative cerebral blood flow (PTA-ROI)	
n	48

Demographics and Baseline characteristics

	all obs
Mean (SD)	1.5 (0.4)
Median	1.4
Min - Max	0.7 - 2.4
Mean relative cerebral blood flow (ROI2)	
n	43
Mean (SD)	1.1 (0.5)
Median	1.1
Min - Max	0.2 - 2.2
Mean relative solid stress (PTA-ROI)	
n	36
Mean (SD)	0.8 (0.1)
Median	0.8
Min - Max	0.6 - 1.1
Mean relative solid stress (ROI2)	
n	29
Mean (SD)	0.9 (0.2)
Median	0.9
Min - Max	0.5 - 1.4
Mean relative apparent diffusion coefficient	
n	46
Mean (SD)	1.4 (0.2)
Median	1.4
Min - Max	1.0 - 1.9
Mean relative fractional anisotropy	
n	46
Mean (SD)	0.5 (0.1)
Median	0.5
Min - Max	0.3 - 1.0
Mean relative restricted spectrum imaging	
n	44
Mean (SD)	0.5 (0.1)
Median	0.5
Min - Max	0.3 - 1.0
Mean DSC K2 leakage	

Demographics and Baseline characteristics

	all obs
n	48
Mean (SD)	0.1 (1.5)
Median	0.4
Min - Max	-3.9 - 2.8
Mean relative cerebral blood	
n	48
Mean (SD)	1.7 (0.5)
Median	1.7
Min - Max	1.0 - 2.8
Mean relative transit time	
n	48
Mean (SD)	1.4 (0.2)
Median	1.3
Min - Max	0.9 - 2.1
Total volume of contrast agent enhancing lesion on T1-weighted MRI in cm3	
n	48
Mean (SD)	6.6 (6.6)
Median	4.6
Min - Max	0.0 - 26.3
Total volume of pathologic signal on T2/FLAIR-weighted MRI in cm3	
n	48
Mean (SD)	13.9 (14.5)
Median	8.8
Min - Max	0.1 - 57.4
Mean tissue elasticity	
n	36
Mean (SD)	0.8 (0.1)
Median	0.8
Min - Max	0.6 - 1.2
Mean tissue relative viscosity	
n	36
Mean (SD)	0.9 (0.2)
Median	0.9
Min - Max	0.0 - 1.1

Demographics and Baseline characteristics

	all obs
Mean tissue stiffness	
n	36
Mean (SD)	0.8 (0.1)
Median	0.8
Min - Max	0.6 - 1.1
Mean tissue viscosity	
n	36
Mean (SD)	0.7 (0.2)
Median	0.7
Min - Max	-0.0 - 1.1
Mean relative DSC extravasation	
n	26
Mean (SD)	1.2 (0.9)
Median	0.9
Min - Max	0.7 - 5.1
Mean relative direction value	
n	26
Mean (SD)	0.5 (1.3)
Median	0.3
Min - Max	-2.7 - 5.4
Mean relative vessel caliber	
n	26
Mean (SD)	1.0 (0.3)
Median	0.9
Min - Max	0.5 - 1.9
Mean relative vessel size index	
n	26
Mean (SD)	1.4 (0.5)
Median	1.1
Min - Max	0.6 - 2.6
Mean relative vortex area	
n	26
Mean (SD)	1.0 (0.1)
Median	1.0

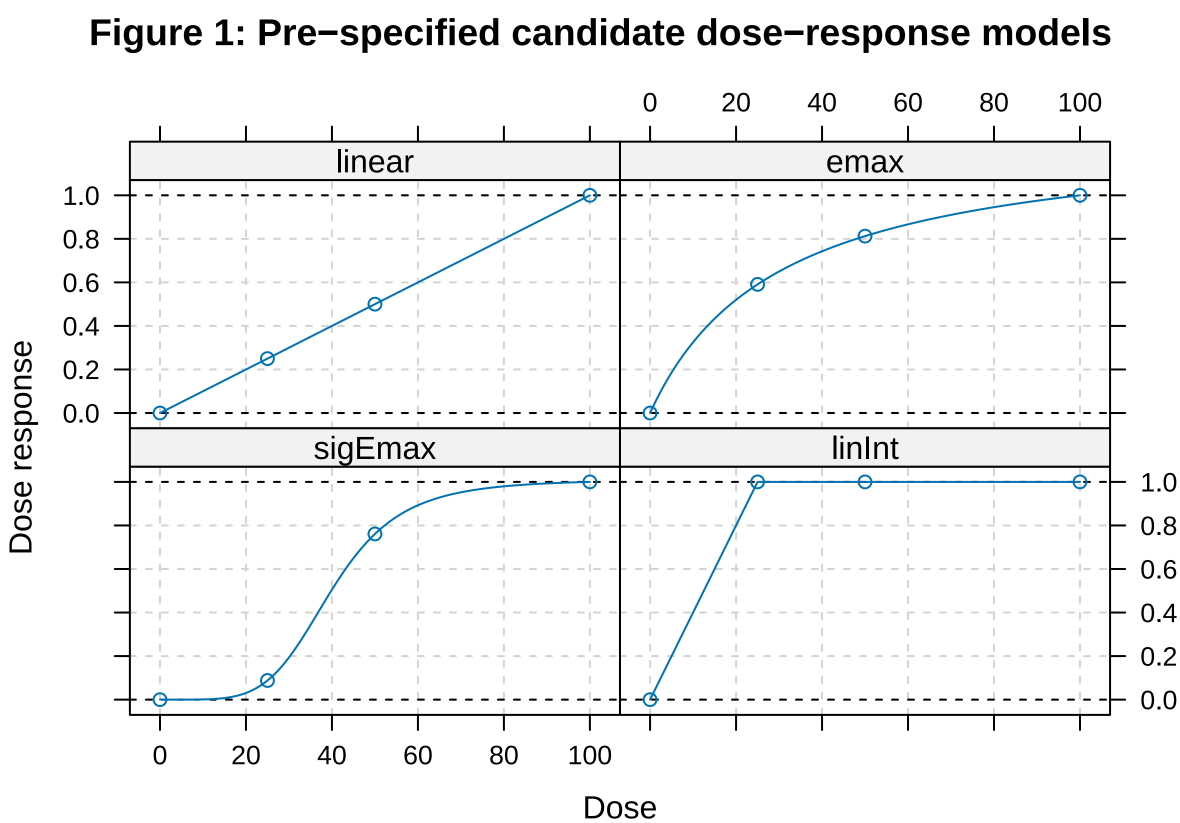
Demographics and Baseline characteristics

	all obs
Min - Max	0.8 - 1.1

Primary Efficacy Analyses (MRI)

The primary variable is analysed with the MCP-Mod methodology as described in the protocol and SAP. We pre-specify four candidate dose-response models (linear, Emax, sigmoid Emax, logistic) and derive optimal contrasts using the estimated variance-covariance from the mixed model. The MCP step tests for a dose-response signal across candidate models with multiplicity control; if significant, the Mod step selects the best-fitting model (by AIC) and reports ED values. Results below are grouped by MRI endpoint (primary, key secondary, sensitivity).

Candidate dose-response models



Below we summarise the MCP-Mod analyses for the four pre-specified MRI endpoints, followed by summary contrasts for the remaining MRI endpoints.

Mean relative cerebral blood flow (PTA-ROI)

Optimal contrasts

Treatment	linear	emax	sigEmax	linInt
0 mg	-0.663	-0.826	-0.592	-0.866
25 mg	-0.202	0.045	-0.390	0.307
50 mg	0.162	0.318	0.405	0.298
100 mg	0.703	0.463	0.577	0.261

MCP-Mod tests

Candidate model	t-statistic	Adjusted p-value
linInt	-1.01	0.93
sigEmax	-1.22	0.96
emax	-1.27	0.96
linear	-1.46	0.98

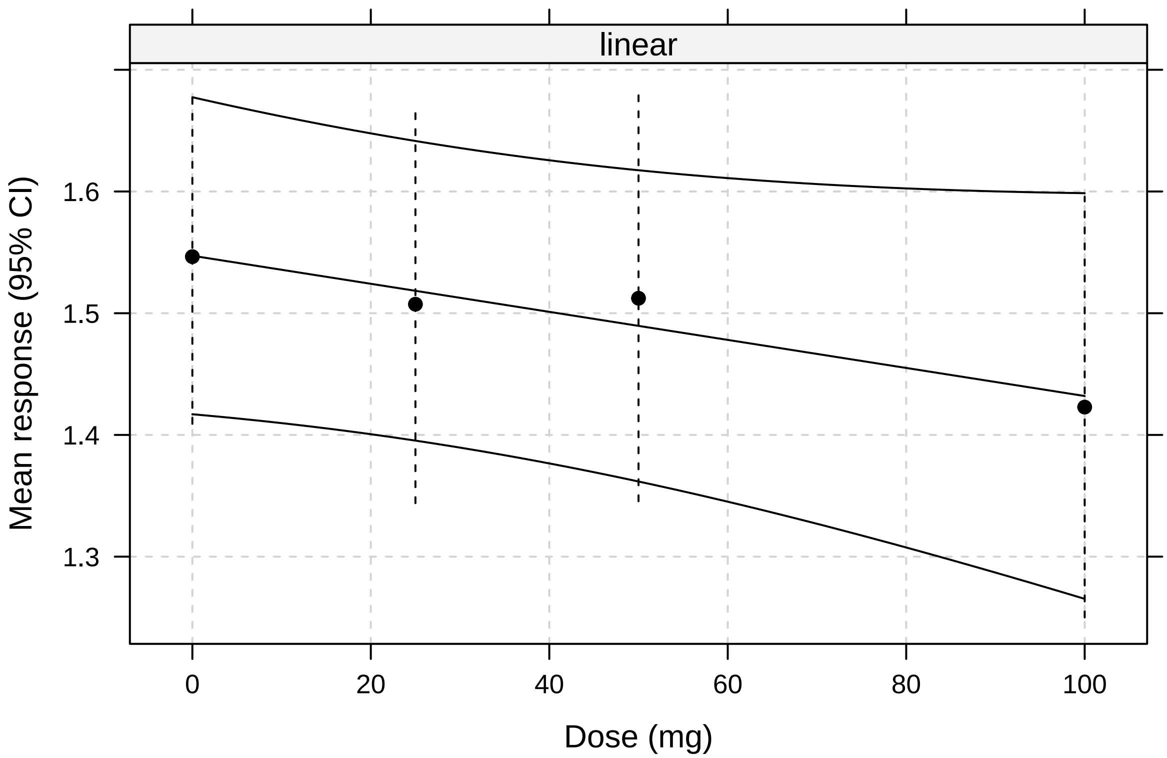
Fitted model parameters (AIC/ED)

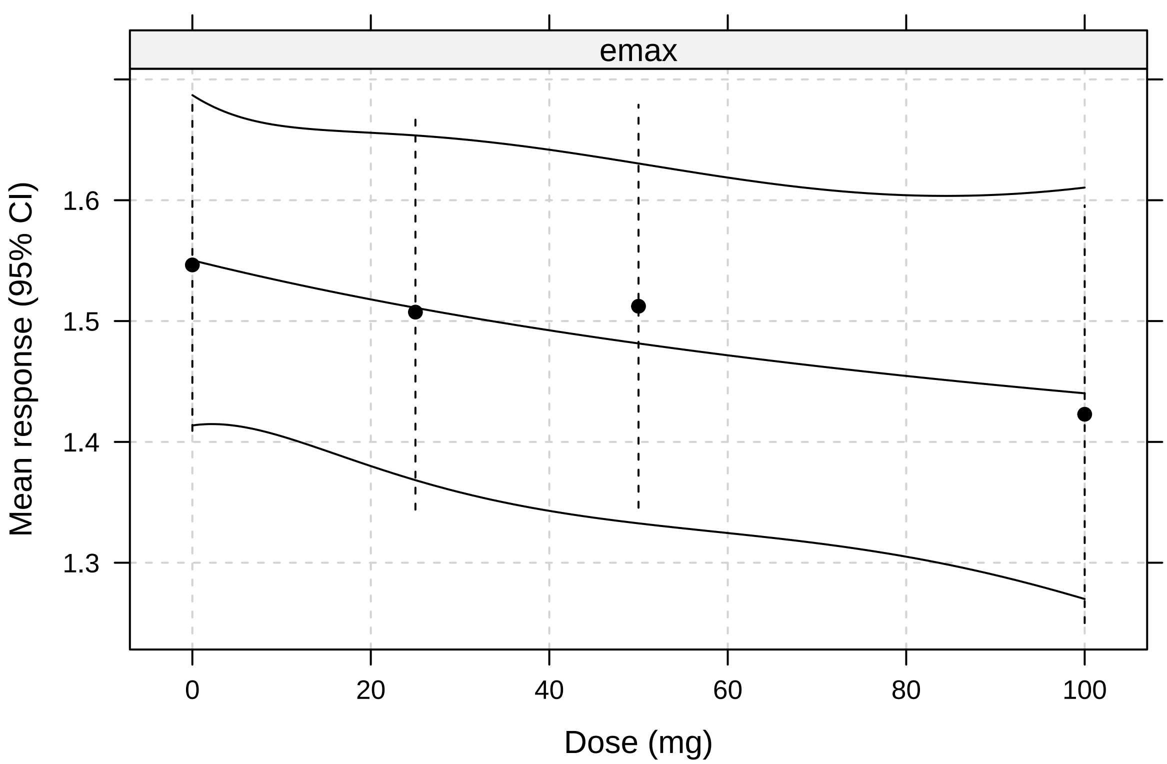
Model	AIC-value	ED_{50}	ED_{70}	ED_{90}
linear	4.20	50.00	70.00	90.00
emax	6.35	37.50	58.33	84.37
linInt	8.00	65.44	79.27	93.09
sigEmax	8.19	63.18	78.06	92.63

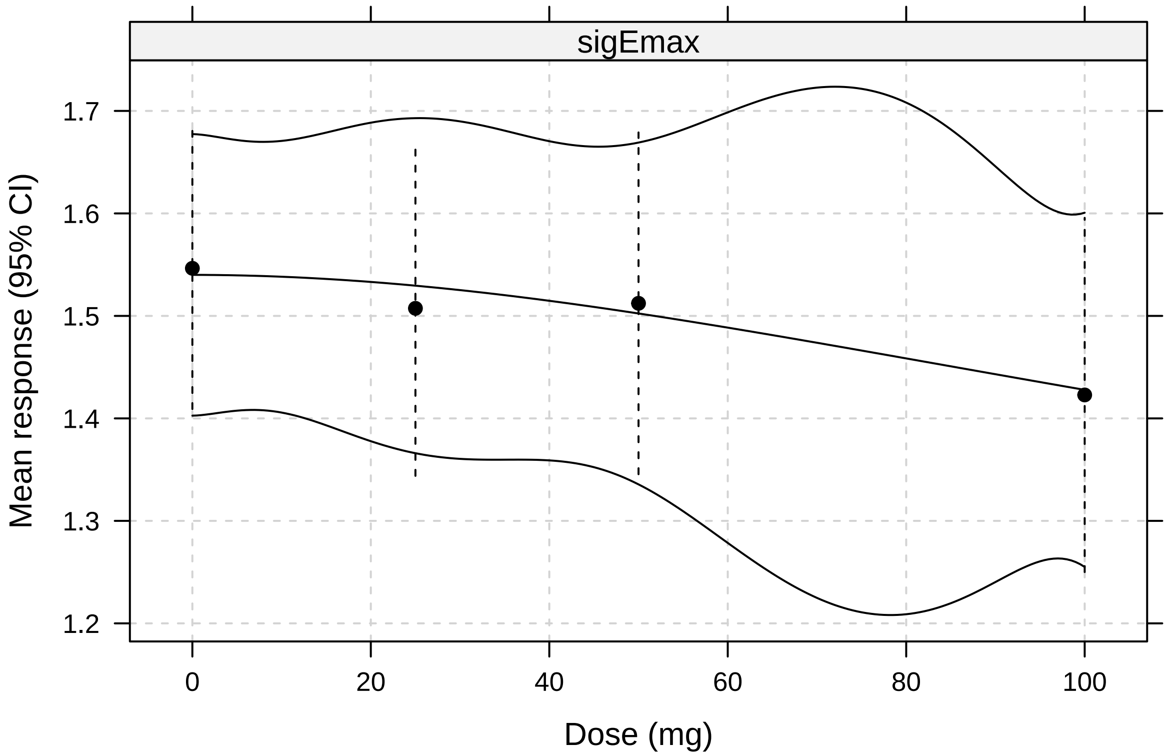
Estimated means by dose

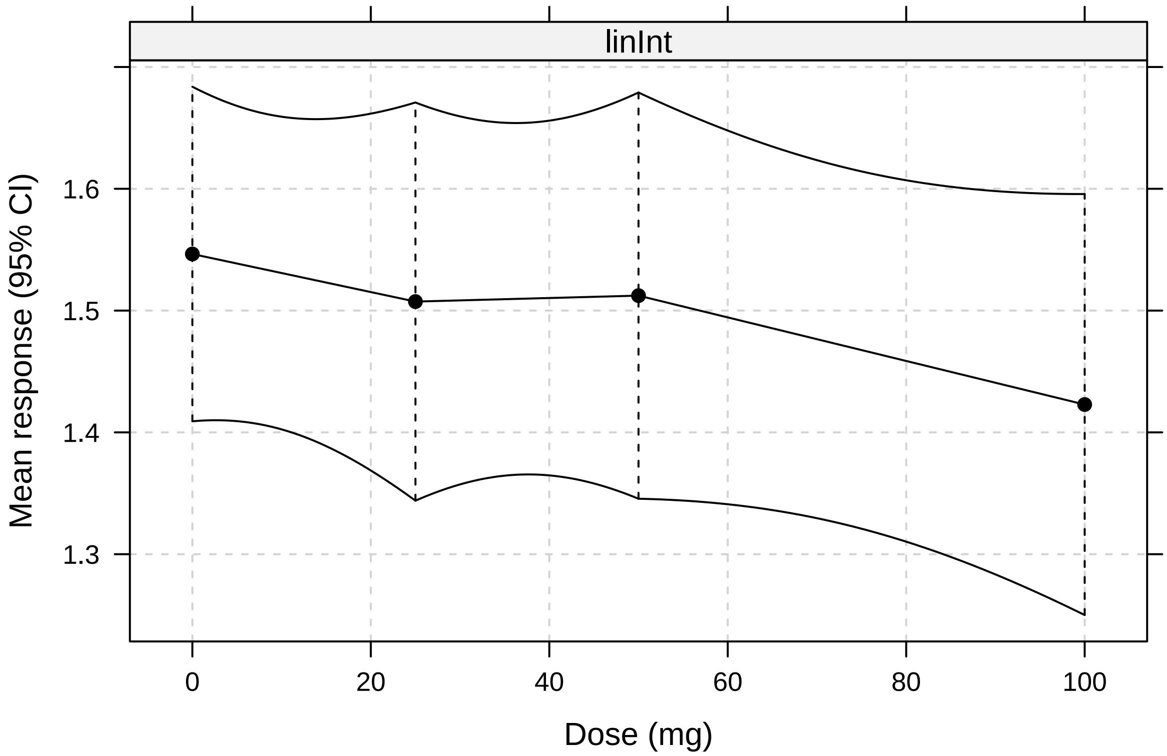
Treatment	Estimated mean (95% CI)
0 mg	1.55 (1.41 to 1.69)
25 mg	1.51 (1.34 to 1.67)
50 mg	1.51 (1.34 to 1.68)
100 mg	1.42 (1.25 to 1.6)

Fitted dose-response curves






sigEmax
Mean response (95% CI)
1.7
1.6
1.5
1.4
1.3
1.2
0
20
40
60
80
100
Dose (mg)



Mean relative cerebral blood flow (ROI2)

Optimal contrasts

Treatment	linear	emax	sigEmax	linInt
0 mg	-0.663	-0.827	-0.582	-0.865
25 mg	-0.210	0.043	-0.406	0.311
50 mg	0.175	0.333	0.429	0.305
100 mg	0.697	0.451	0.559	0.248

MCP-Mod tests

Candidate model	t-statistic	Adjusted p-value
sigEmax	-0.70	0.88
linear	-0.95	0.92
emax	-1.15	0.95
linInt	-1.24	0.96

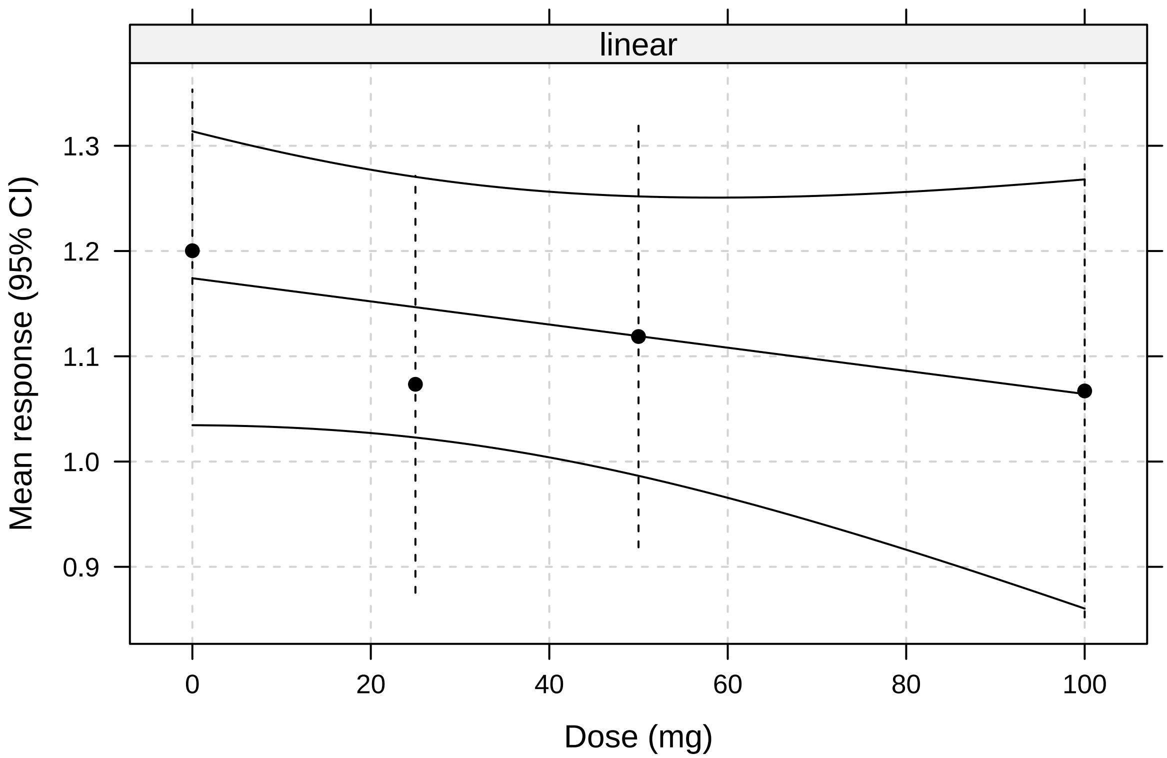
Fitted model parameters (AIC/ED)

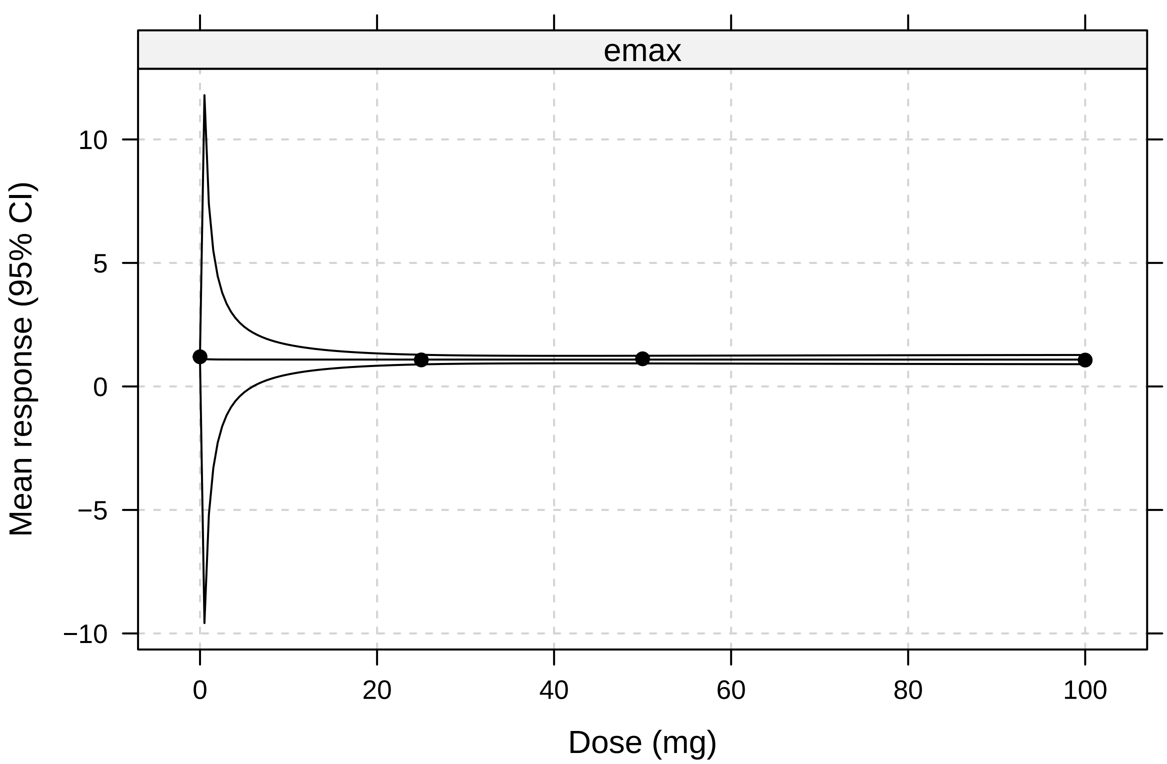
Model	AIC-value	ED_{50}	ED_{70}	ED_{90}
linear	4.86	50.00	70.00	90.00
emax	6.22	0.10	0.23	0.89
linInt	8.00	13.12	18.37	23.62
sigEmax	8.22	0.62	0.71	0.88

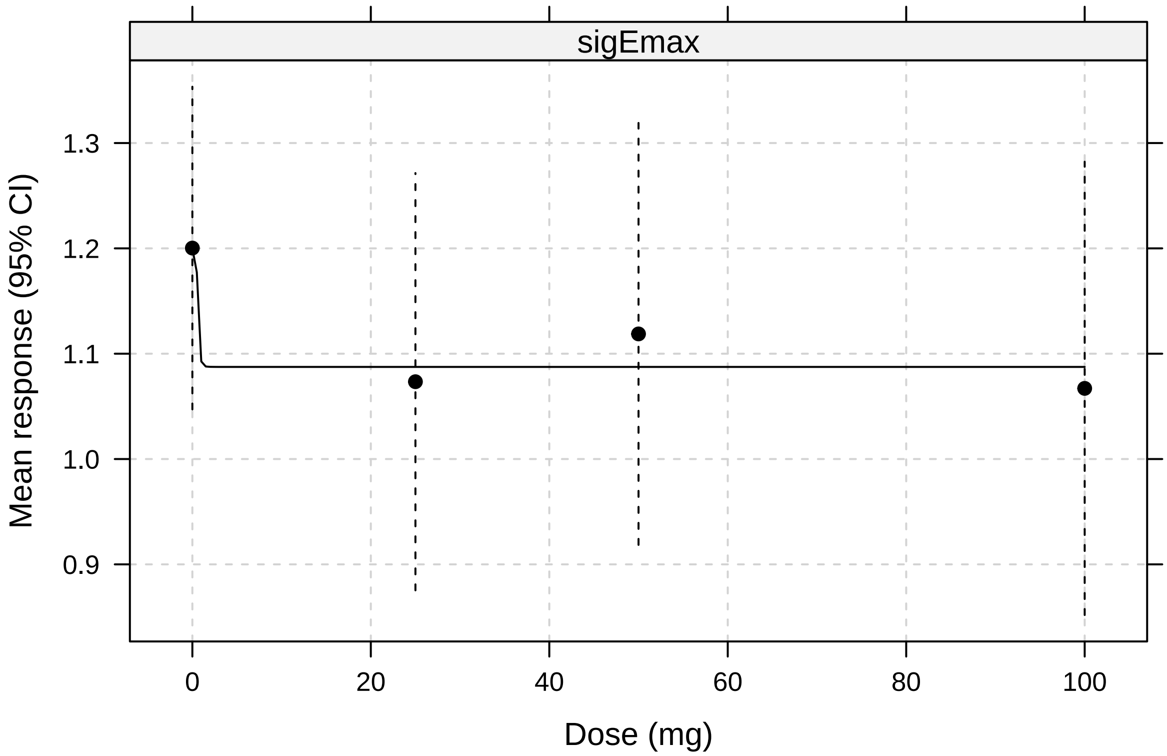
Estimated means by dose

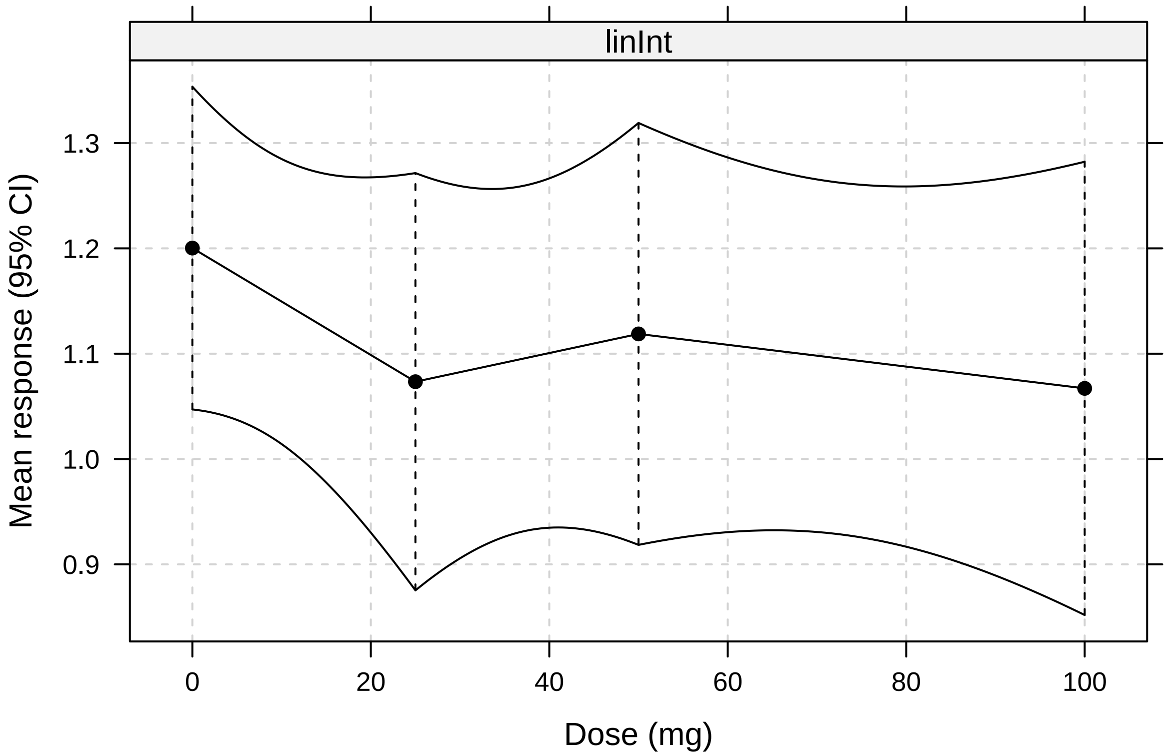
Treatment	Estimated mean (95% CI)
0 mg	1.2 (1.04 to 1.36)
25 mg	1.07 (0.87 to 1.27)
50 mg	1.12 (0.92 to 1.32)
100 mg	1.07 (0.85 to 1.28)

Fitted dose-response curves






sigEmax
Mean response (95% CI)
1.3
1.2
1.1
1.0
0.9
0
20
40
60
80
100
Dose (mg)



Mean relative solid stress (PTA-ROI)

Optimal contrasts

Treatment	linear	emax	sigEmax	linInt
0 mg	-0.621	-0.805	-0.580	-0.865
25 mg	-0.231	-0.010	-0.395	0.256
50 mg	0.112	0.305	0.361	0.319
100 mg	0.740	0.509	0.614	0.289

MCP-Mod tests

Candidate model	t-statistic	Adjusted p-value
linear	-1.21	0.95
sigEmax	-1.52	0.98
emax	-1.71	0.99
linInt	-1.74	0.99

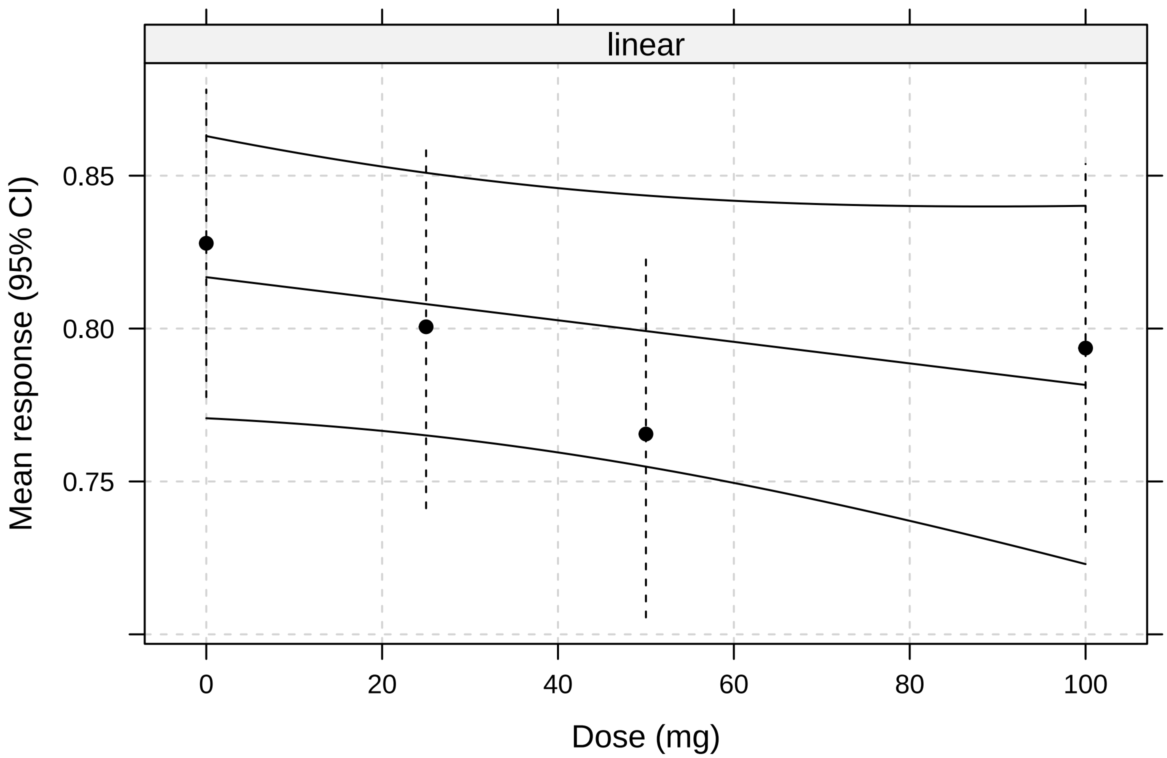
Fitted model parameters (AIC/ED)

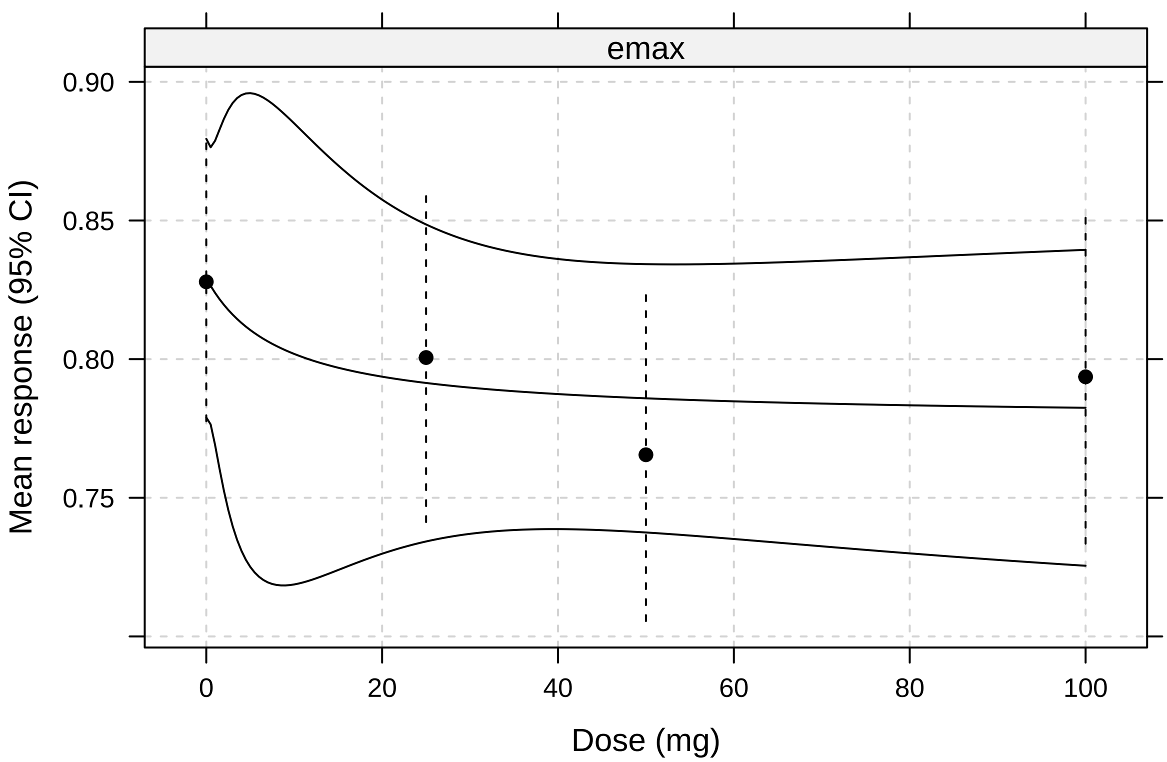
Model	AIC-value	ED_{50}	ED_{70}	ED_{90}
linear	6.95	50.00	70.00	90.00
emax	7.27	7.32	15.56	41.54
linInt	8.00	27.76	36.66	45.55
sigEmax	8.82	24.27	26.42	30.23

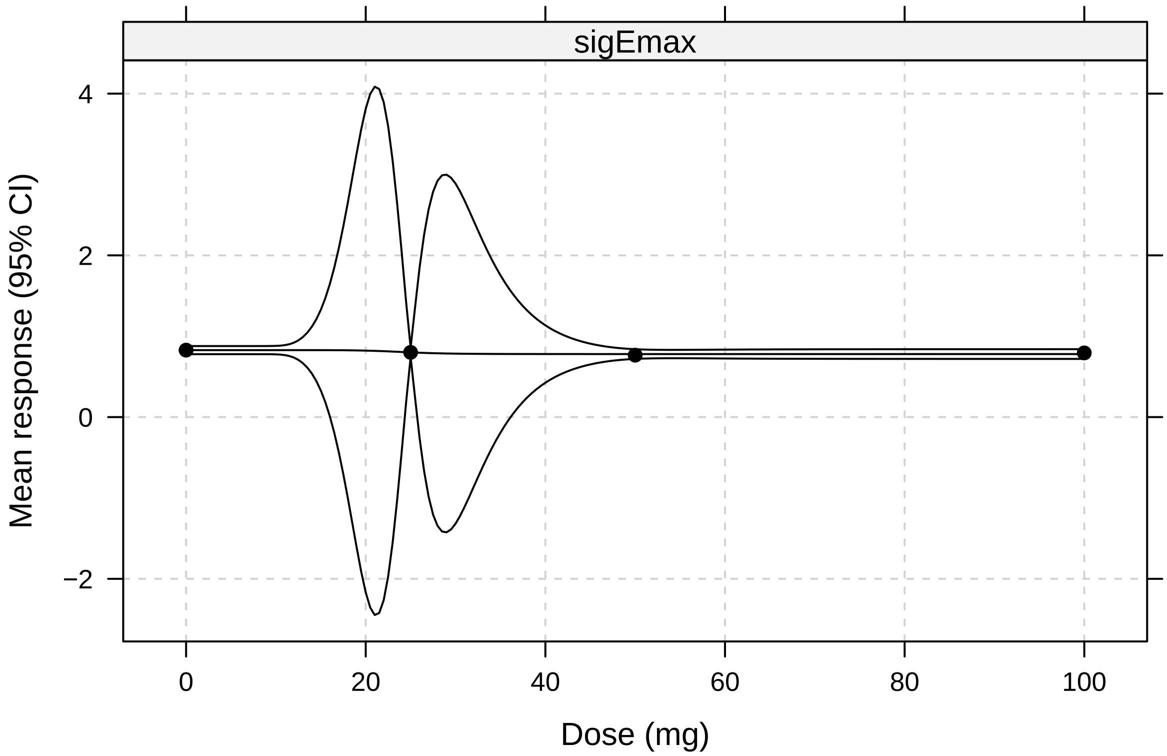
Estimated means by dose

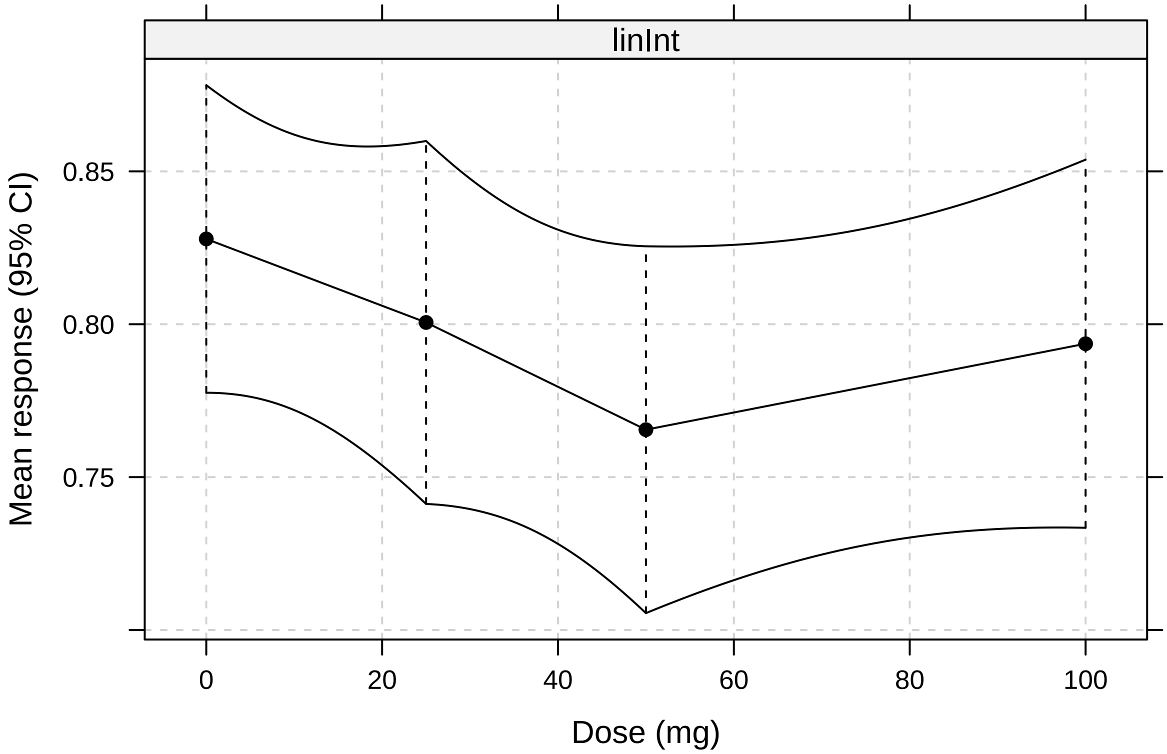
Treatment	Estimated mean (95% CI)
0 mg	0.83 (0.78 to 0.88)
25 mg	0.8 (0.74 to 0.86)
50 mg	0.77 (0.7 to 0.83)
100 mg	0.79 (0.73 to 0.85)

Fitted dose-response curves






sigEmax
Mean response (95% CI)
4
2
0
−2
0
20
40
60
80
100
Dose (mg)



Mean relative solid stress (ROI2)

Optimal contrasts

Treatment	linear	emax	sigEmax	linInt
0 mg	-0.579	-0.797	-0.525	-0.866
25 mg	-0.272	-0.022	-0.454	0.293
50 mg	0.086	0.288	0.351	0.295
100 mg	0.764	0.531	0.629	0.277

MCP-Mod tests

Candidate model	t-statistic	Adjusted p-value
linear	-2.59	1
sigEmax	-2.86	1
linInt	-3.06	1
emax	-3.22	1

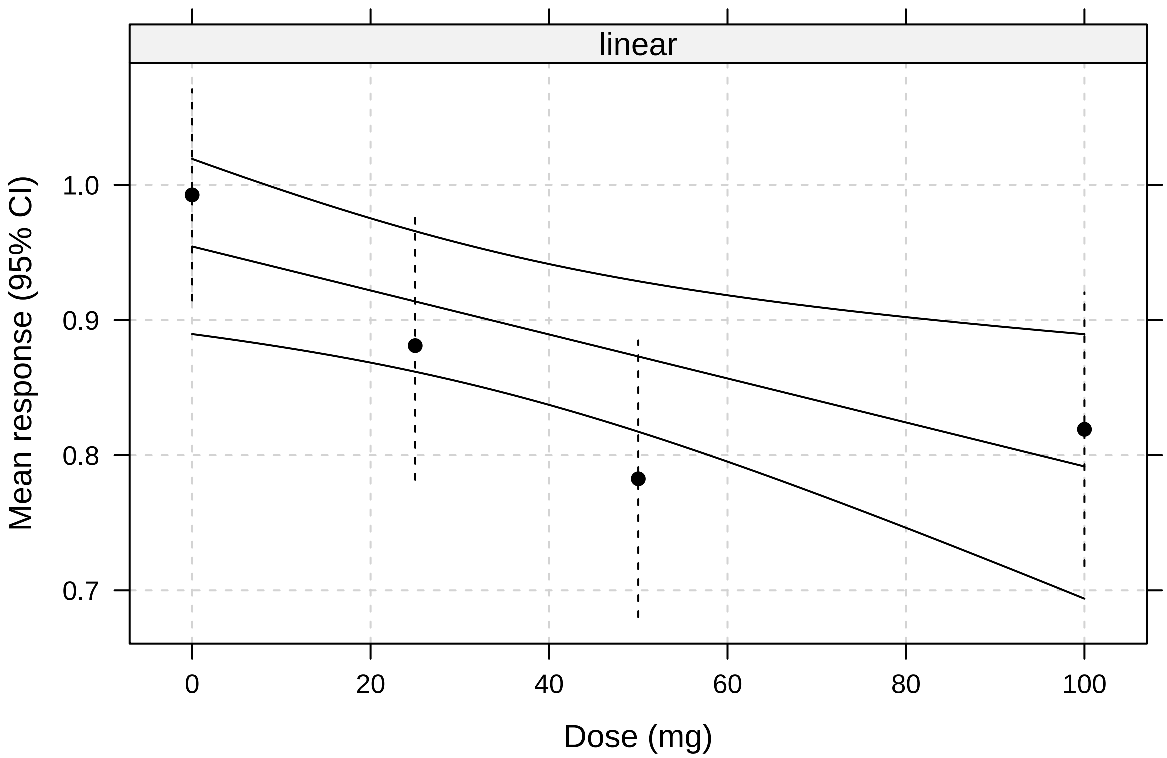
Fitted model parameters (AIC/ED)

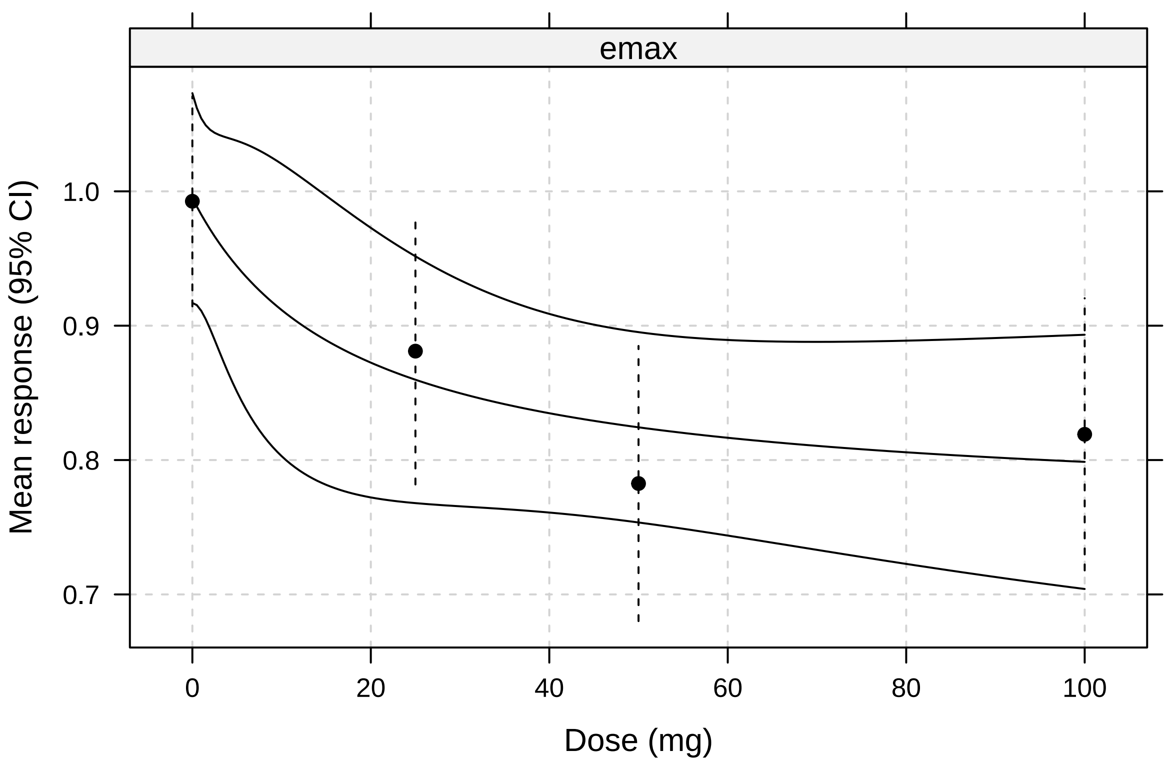
Model	AIC-value	ED_{50}	ED_{70}	ED_{90}
emax	7.23	13.10	26.02	57.56
linInt	8.00	23.54	34.01	44.67
sigEmax	8.32	24.17	26.31	30.11
linear	9.02	50.00	70.00	90.00

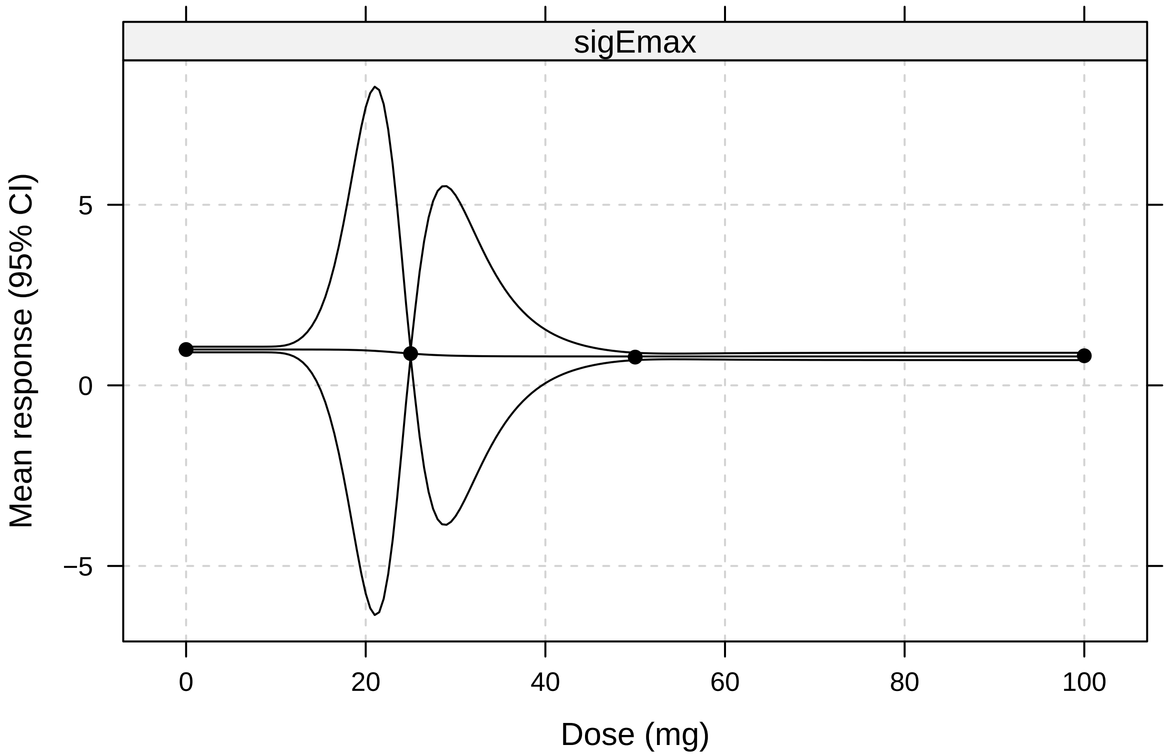
Estimated means by dose

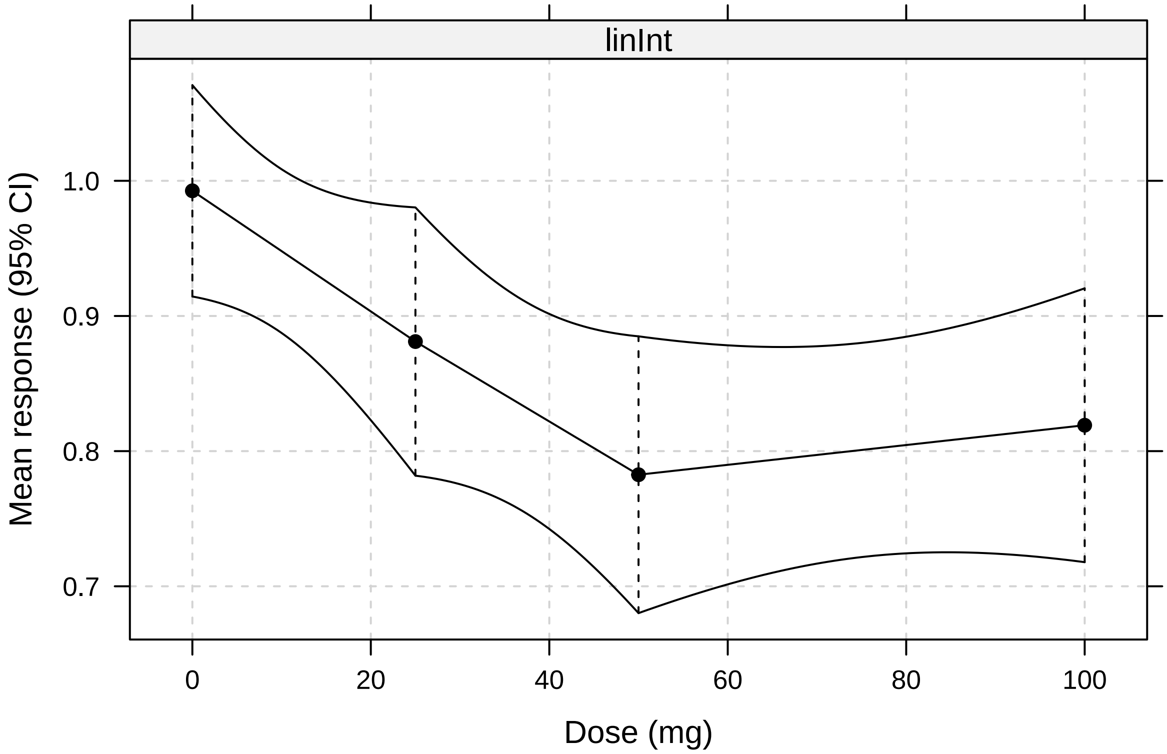
Treatment	Estimated mean (95% CI)
0 mg	0.99 (0.91 to 1.07)
25 mg	0.88 (0.78 to 0.98)
50 mg	0.78 (0.68 to 0.89)
100 mg	0.82 (0.72 to 0.92)

Fitted dose-response curves






sigEmax
Mean response (95% CI)
5
0
−5
0
20
40
60
80
100
Dose (mg)



Other MRI Endpoints

For remaining MRI endpoints, we present estimated means by dose, contrasts vs 0 mg, and a plot with confidence intervals.

mrmef

Means by dose

A tibble: 4 x 10

trtcd estimate std.error df conf.low conf.high statistic p.value trt
1 0 -0.296 0.248 80.0 -0.789 0.197 -1.20 0.235 0 mg
2 25 -0.0152 0.309 128. -0.627 0.597 -0.0491 0.961 25 mg 3 50 0.146 0.317 133. -0.481 0.773 0.460 0.646 50 mg 4 100 0.0546 0.331 139. -0.599 0.709 0.165 0.869 100 mg # i 1 more variable: mean_txt

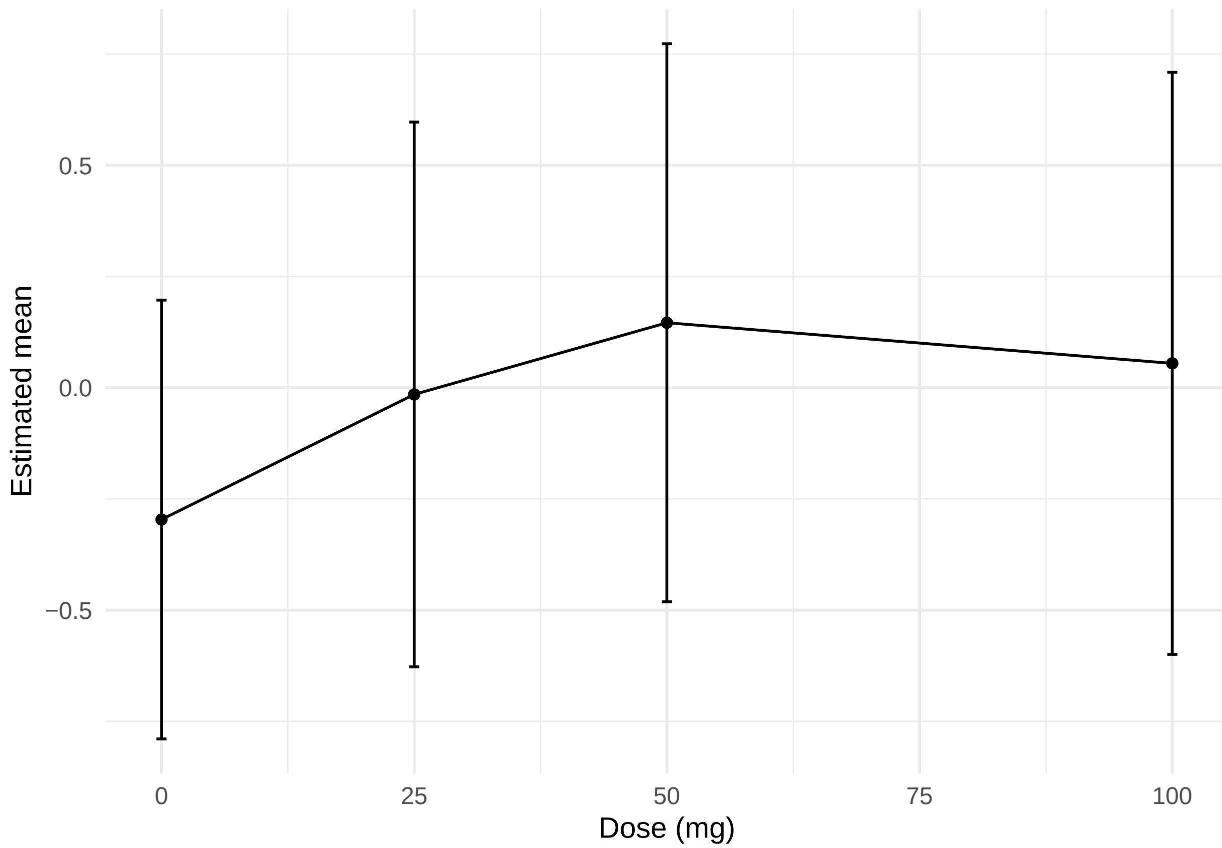
Differences vs 0 mg

A tibble: 3 x 11

term contrast null.value estimate std.error df conf.low conf.high 1 trtcd 25 mg vs 0 mg 0 0.281 0.320 138. -0.482 1.04 2 trtcd 50 mg vs 0 mg 0 0.442 0.325 137. -0.333 1.22 3 trtcd 100 mg vs 0 mg 0 0.351 0.338 139.

-0.456 1.16 # i 3 more variables: statistic , adj.p.value , estimate_txt

Plot



mrdsc

Means by dose

A tibble: 4 x 10

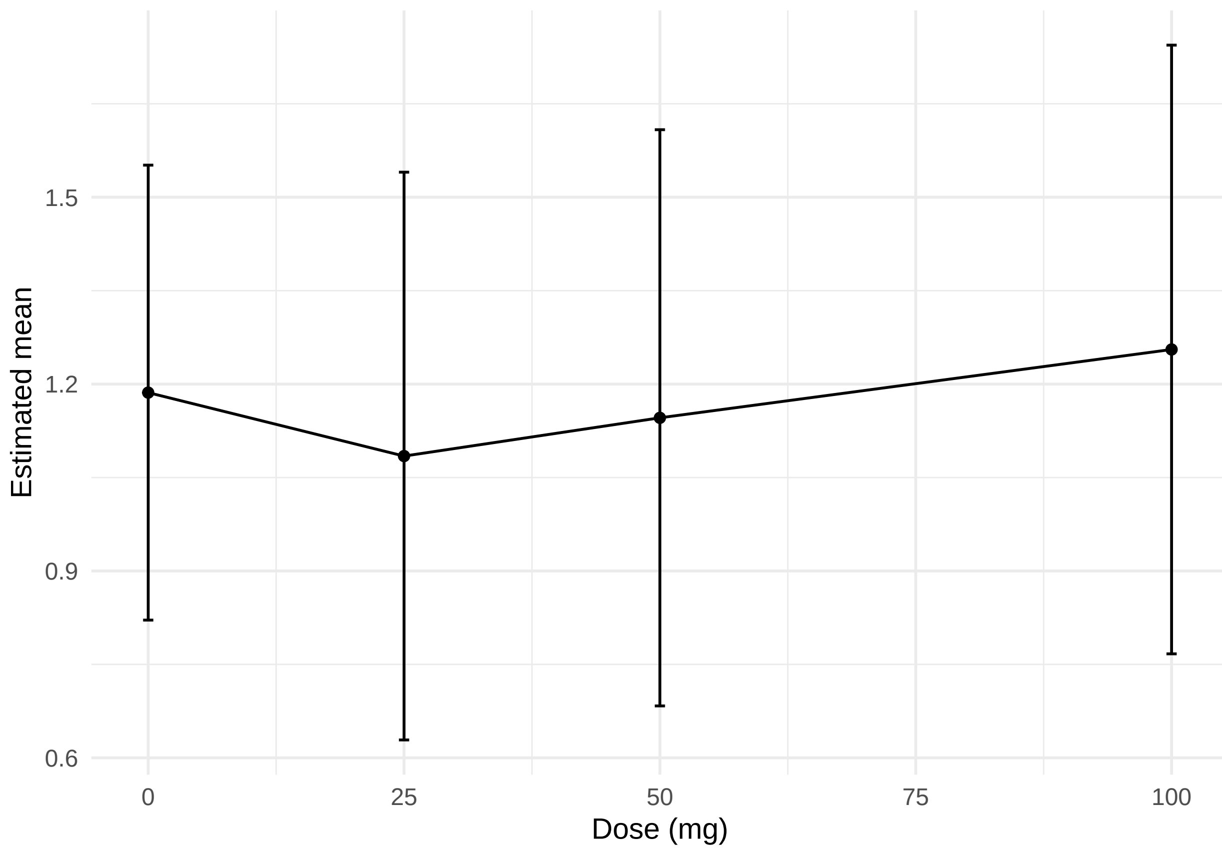
trtcd estimate std.error df conf.low conf.high statistic p.value trt
1 0 1.19 0.182 48.4 0.821 1.55 6.53 0.0000000374 0 mg 2 25 1.08 0.229 72.6 0.629 1.54 4.74 0.0000103 25 mg 3 50 1.15 0.232 75.1 0.683 1.61 4.93 0.00000470 50 mg 4 100 1.26 0.246 79.6 0.767 1.74 5.11 0.00000213 100 ~ # i 1 more variable: mean_txt

Differences vs 0 mg

A tibble: 3 x 11

term contrast null.value estimate std.error df conf.low conf.high 1 trtcd 25 mg vs 0 mg 0 -0.102 0.243 73.4 -0.687 0.483 2 trtcd 50 mg vs 0 mg 0 -0.0404 0.243 71.5 -0.626 0.545 3 trtcd 100 mg vs 0 mg 0 0.0693 0.254 71.6 -0.543 0.681 # i 3 more variables: statistic , adj.p.value , estimate_txt

Plot



mrdpta

Means by dose

A tibble: 4 x 10

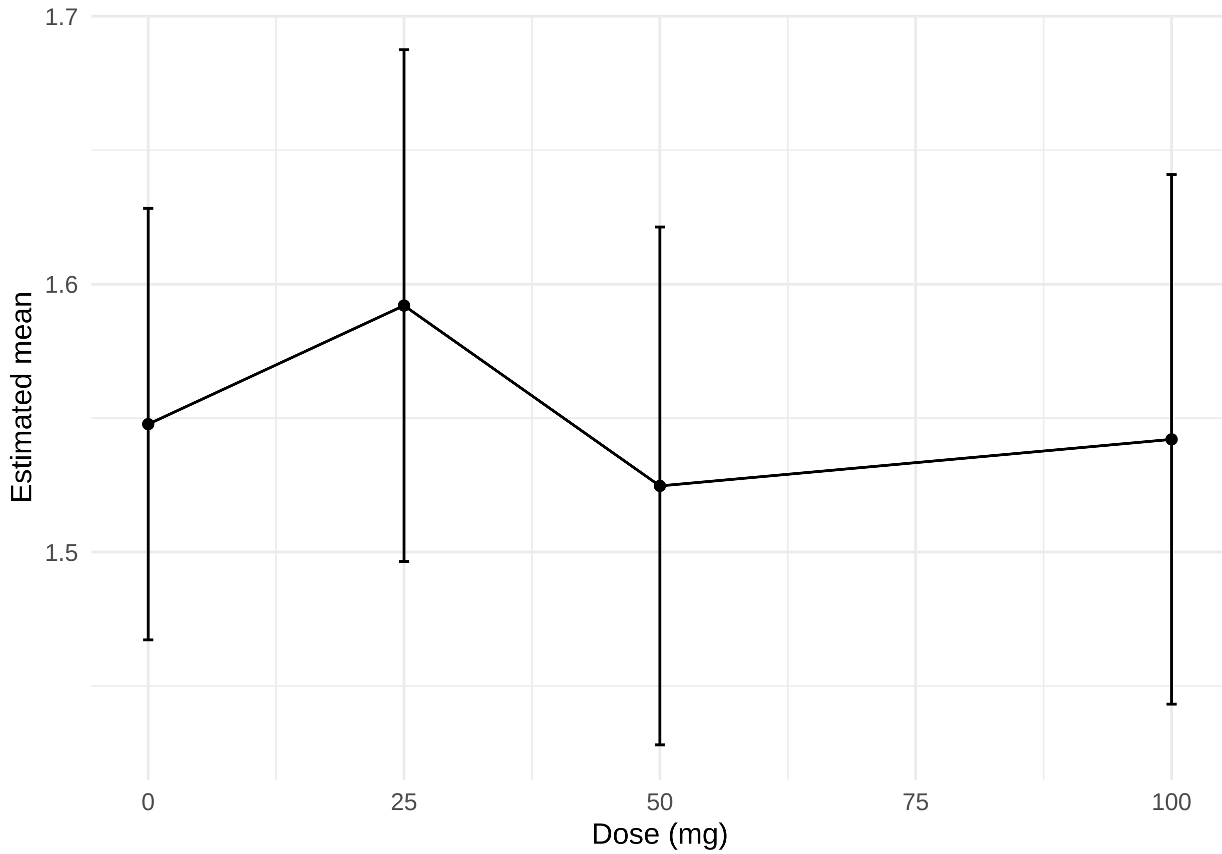
trtcd estimate std.error df conf.low conf.high statistic p.value trt
1 0 1.55 0.0404 71.1 1.47 1.63 38.3 3.30e-49 0 mg
2 25 1.59 0.0482 112. 1.50 1.69 33.0 1.05e-59 25 mg 3 50 1.52 0.0488 116. 1.43 1.62 31.2 2.51e-58 50 mg 4 100 1.54 0.0499 120. 1.44 1.64 30.9 5.03e-59 100 mg # i 1 more variable: mean_txt

Differences vs 0 mg

A tibble: 3 x 11

term contrast null.value estimate std.error df conf.low conf.high 1 trtcd 25 mg vs 0 mg 0 0.0443 0.0451 134. -0.0633 0.152 2 trtcd 50 mg vs 0 mg 0 -0.0230 0.0455 133. -0.132 0.0857 3 trtcd 100 mg vs 0 mg 0 -0.00567 0.0468 135. -0.117 0.106 # i 3 more variables: statistic , adj.p.value , estimate_txt

Plot



mrpta

Means by dose

A tibble: 4 x 10

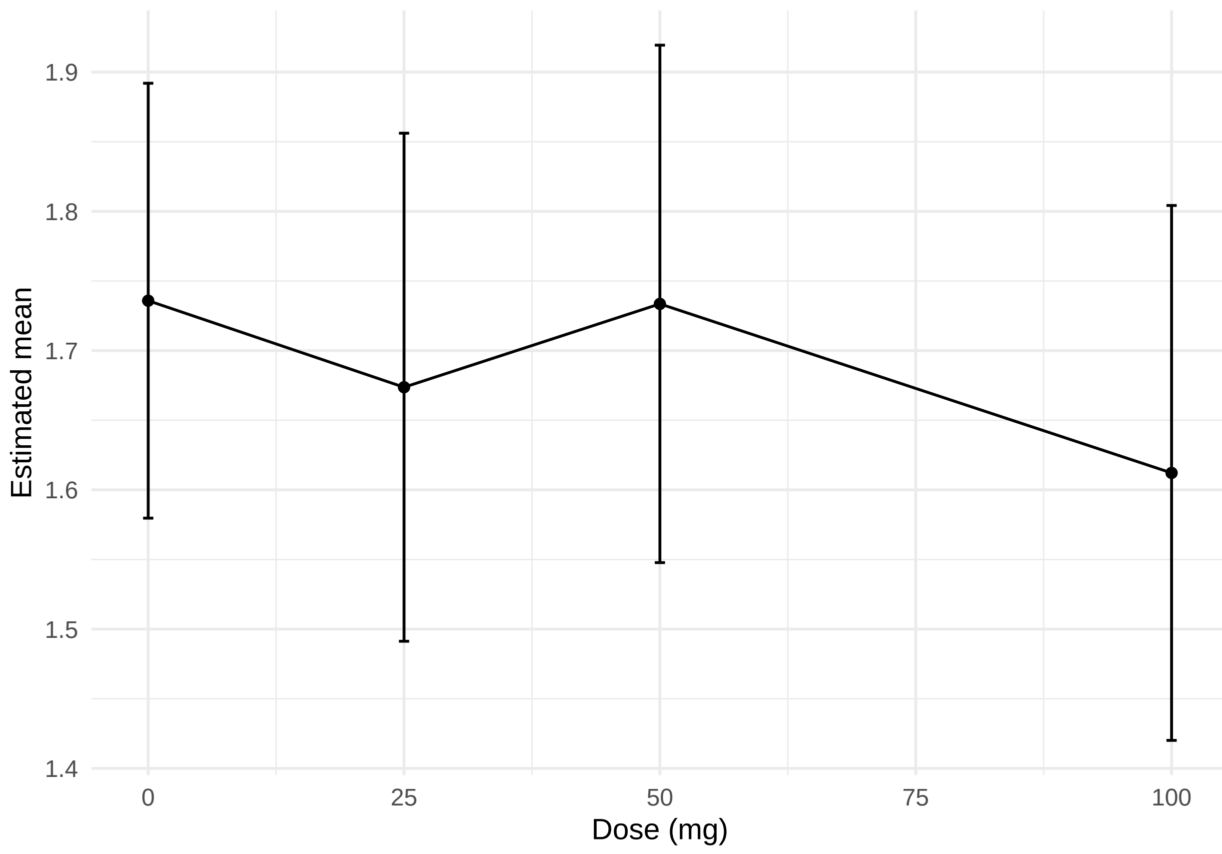
trtcd estimate std.error df conf.low conf.high statistic p.value trt
1 0 1.74 0.0782 67.2 1.58 1.89 22.2 1.19e-32 0 mg
2 25 1.67 0.0920 106. 1.49 1.86 18.2 2.34e-34 25 mg 3 50 1.73 0.0938 111. 1.55 1.92 18.5 1.28e-35 50 mg 4 100 1.61 0.0970 118. 1.42 1.80 16.6 9.96e-33 100 mg # i 1 more variable: mean_txt

Differences vs 0 mg

A tibble: 3 x 11

term contrast null.value estimate std.error df conf.low conf.high 1 trtcd 25 mg vs 0 mg 0 -0.0621 0.0829 130. -0.260 0.136 2 trtcd 50 mg vs 0 mg 0 -0.00230 0.0841 130. -0.203 0.198 3 trtcd 100 mg vs 0 mg 0 -0.124 0.0876 131. -0.333 0.0856 # i 3 more variables: statistic , adj.p.value , estimate_txt

Plot



mrrdv

Means by dose

A tibble: 4 x 10

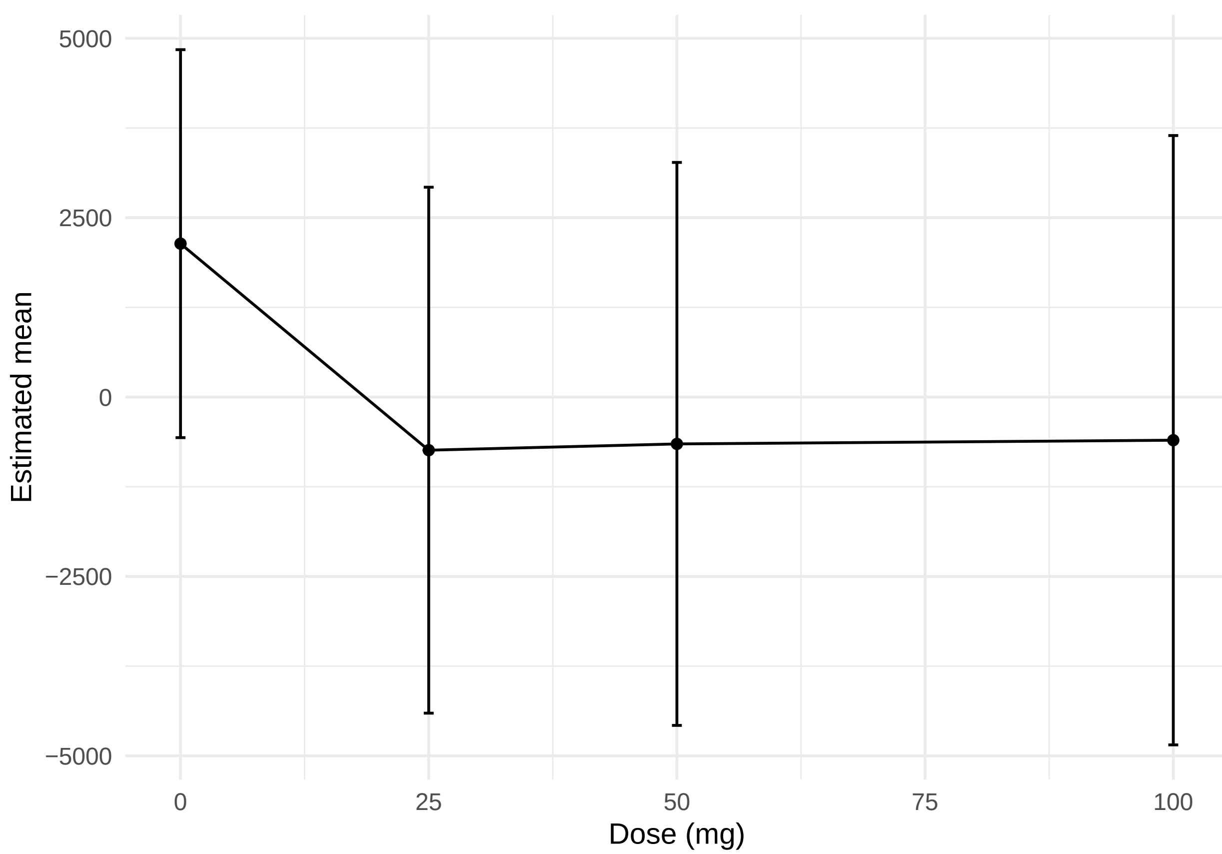
trtcd estimate std.error df conf.low conf.high statistic p.value trt
1 0 2139. 1352. 61.4 -565. 4843. 1.58 0.119 0 mg
2 25 -740. 1834. 62.2 -4406. 2925. -0.404 0.688 25 mg 3 50 -653. 1963. 62.9 -4577. 3271. -0.333 0.741 50 mg 4 100 -601. 2123. 60.3 -4848. 3646. -0.283 0.778 100 mg # i 1 more variable: mean_txt

Differences vs 0 mg

A tibble: 3 x 11

term contrast null.value estimate std.error df conf.low conf.high 1 trtcd 25 mg vs 0 mg 0 -2879. 2577. 79.8 -9080. 3322. 2 trtcd 50 mg vs 0 mg 0 -2792. 2671. 80.2 -9219. 3636. 3 trtcd 100 mg vs 0 mg 0 -2740. 2774. 76.4 -9421. 3941. # i 3 more variables: statistic , adj.p.value , estimate_txt

Plot



mrdrfa

Means by dose

A tibble: 4 x 10

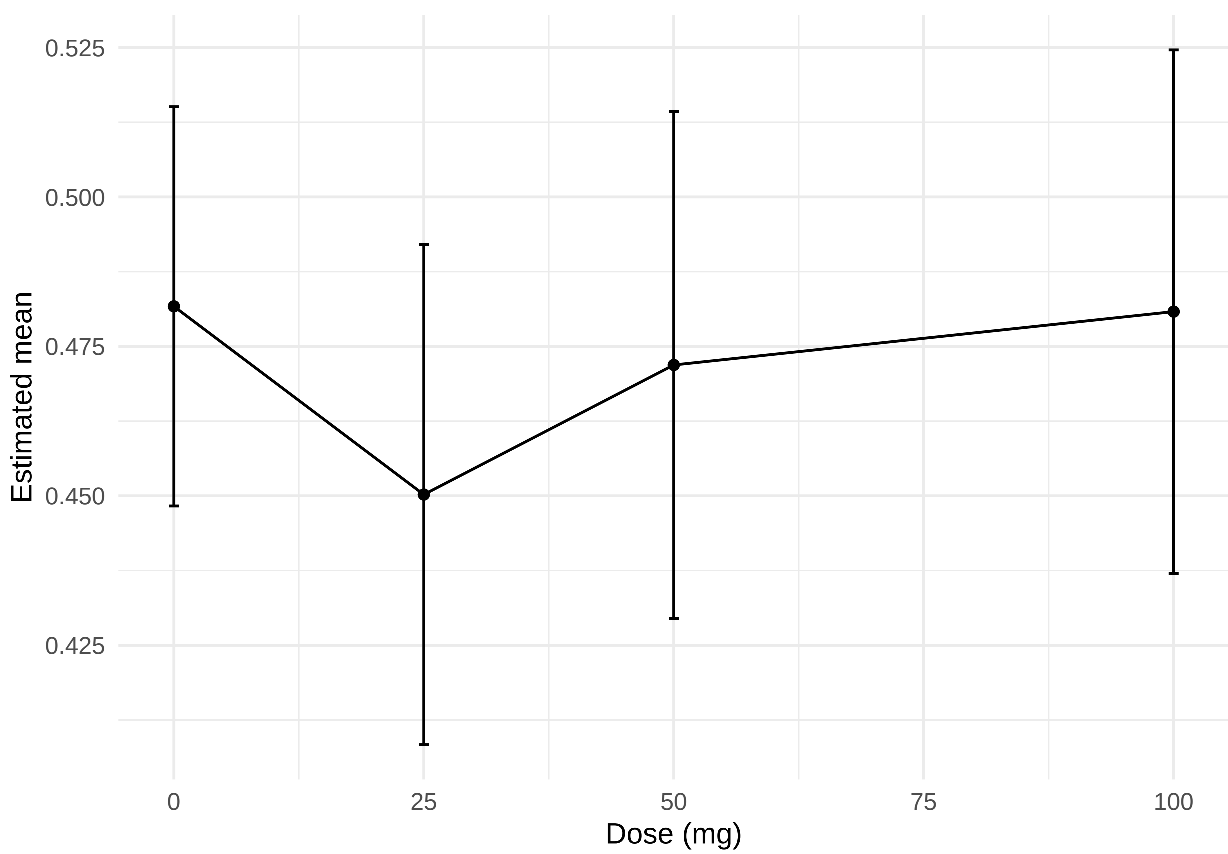
trtcd estimate std.error df conf.low conf.high statistic p.value trt
1 0 0.482 0.0168 81.6 0.448 0.515 28.7 2.08e-44 0 mg
2 25 0.450 0.0212 130. 0.408 0.492 21.3 3.62e-44 25 mg 3 50 0.472 0.0214 133. 0.430 0.514 22.0 3.16e-46 50 mg 4 100 0.481 0.0221 137. 0.437 0.525 21.7 3.38e-46 100 mg # i 1 more variable: mean_txt

Differences vs 0 mg

A tibble: 3 x 11

term contrast null.value estimate std.error df conf.low conf.high 1 trtcd 25 mg vs 0 mg 0 -0.0315 0.0219 138. -0.0837 0.0208 2 trtcd 50 mg vs 0 mg 0 -0.00981 0.0222 138. -0.0627 0.0431 3 trtcd 100 mg vs 0 mg 0 -0.000884 0.0229 140. -0.0554 0.0537 # i 3 more variables: statistic , adj.p.value , estimate_txt

Plot



mrdrsi

Means by dose

A tibble: 4 x 10

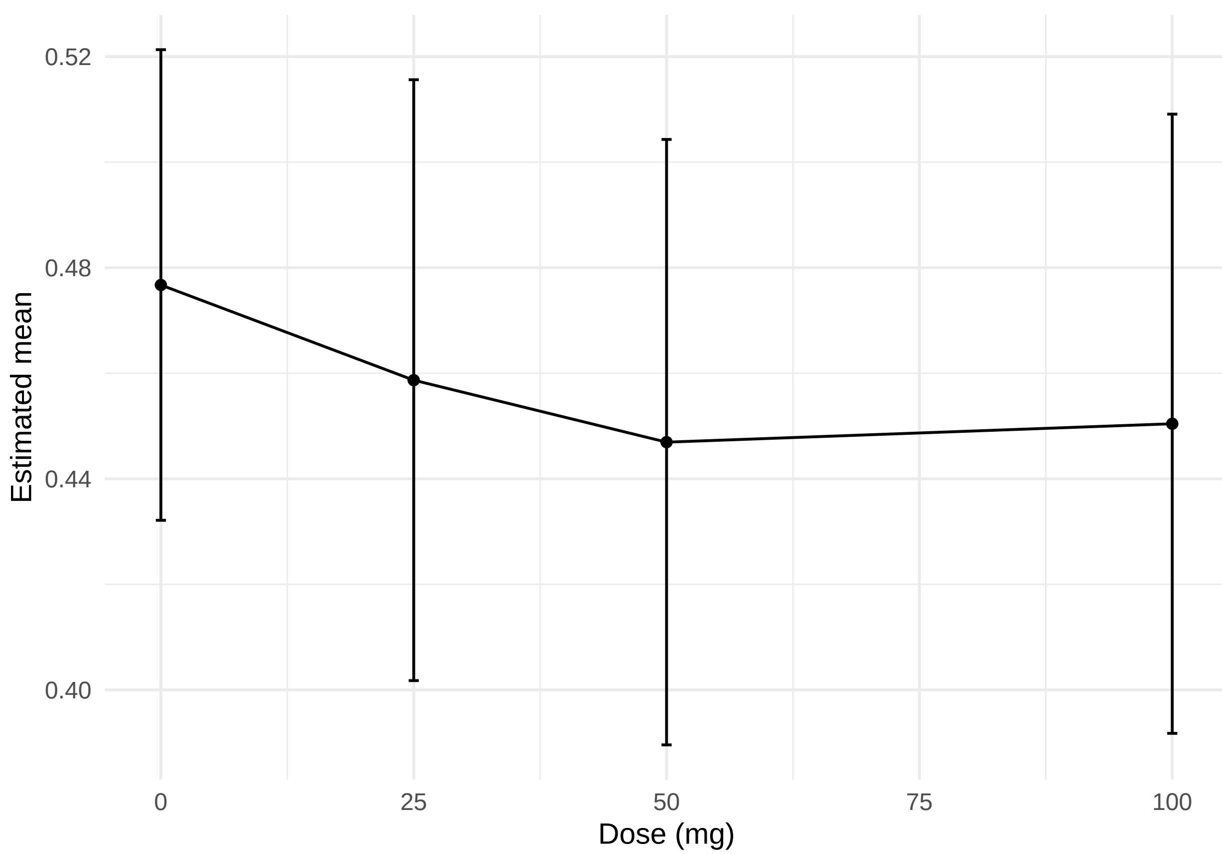
trtcd estimate std.error df conf.low conf.high statistic p.value trt
1 0 0.477 0.0224 86.1 0.432 0.521 21.3 5.31e-36 0 mg
2 25 0.459 0.0288 132. 0.402 0.516 15.9 1.58e-32 25 mg 3 50 0.447 0.0290 137. 0.390 0.504 15.4 1.09e-31 50 mg 4 100 0.450 0.0297 137. 0.392 0.509 15.2 3.68e-31 100 mg # i 1 more variable: mean_txt

Differences vs 0 mg

A tibble: 3 x 11

term contrast null.value estimate std.error df conf.low conf.high 1 trtcd 25 mg vs 0 mg 0 -0.0180 0.0311 141. -0.0923 0.0562 2 trtcd 50 mg vs 0 mg 0 -0.0298 0.0306 135. -0.103 0.0434 3 trtcd 100 mg vs 0 mg 0 -0.0263 0.0316 138. -0.102 0.0492 # i 3 more variables: statistic , adj.p.value , estimate_txt

Plot



mrrtt

Means by dose

A tibble: 4 x 10

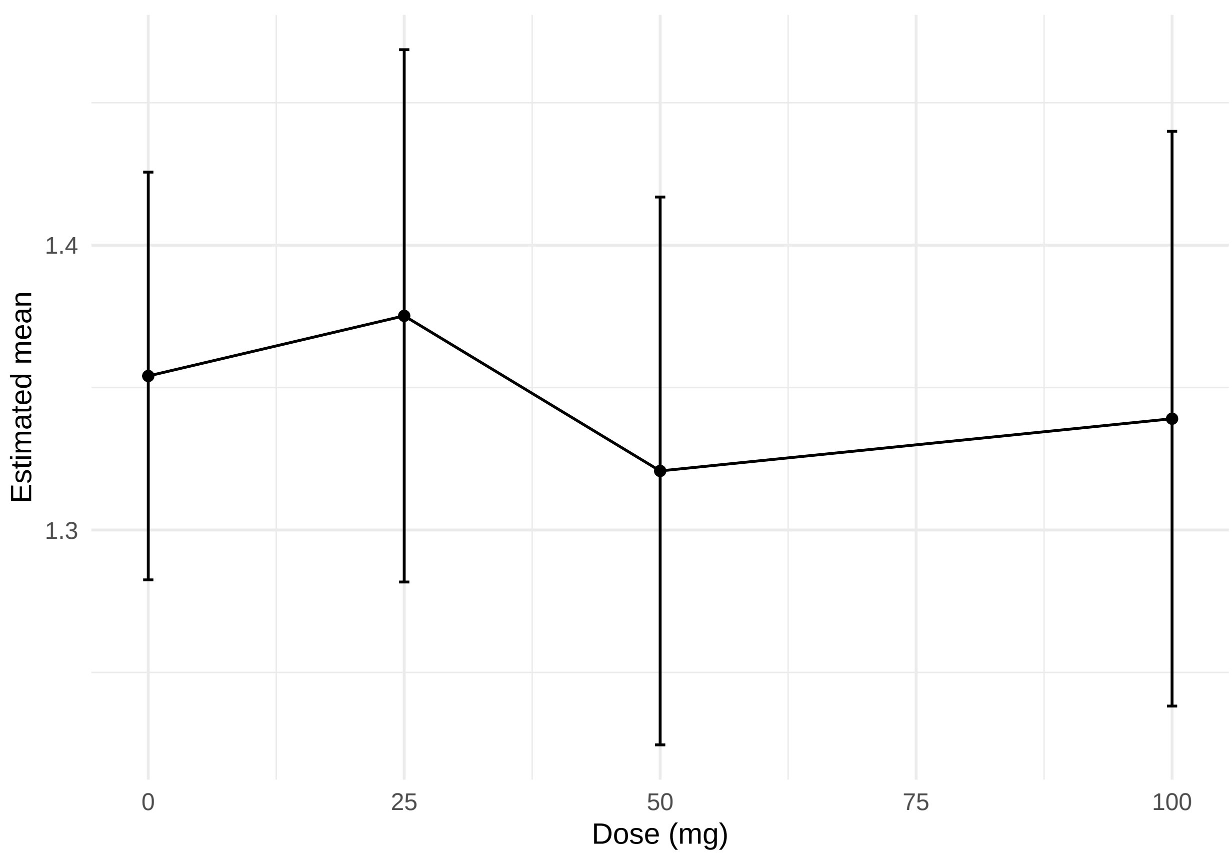
trtcd estimate std.error df conf.low conf.high statistic p.value trt
1 0 1.35 0.0360 93.7 1.28 1.43 37.6 2.76e-58 0 mg
2 25 1.38 0.0473 140. 1.28 1.47 29.1 3.41e-61 25 mg 3 50 1.32 0.0487 144. 1.22 1.42 27.1 1.64e-58 50 mg 4 100 1.34 0.0511 148. 1.24 1.44 26.2 1.56e-57 100 mg # i 1 more variable: mean_txt

Differences vs 0 mg

A tibble: 3 x 11

term contrast null.value estimate std.error df conf.low conf.high 1 trtcd 25 mg vs 0 mg 0 0.0211 0.0534 146. -0.106 0.148 2 trtcd 50 mg vs 0 mg 0 -0.0333 0.0542 146. -0.163 0.0960 3 trtcd 100 mg vs 0 mg 0 -0.0150 0.0563 148. -0.149 0.119 # i 3 more variables: statistic , adj.p.value , estimate_txt

Plot



mrrvc

Means by dose

A tibble: 4 x 10

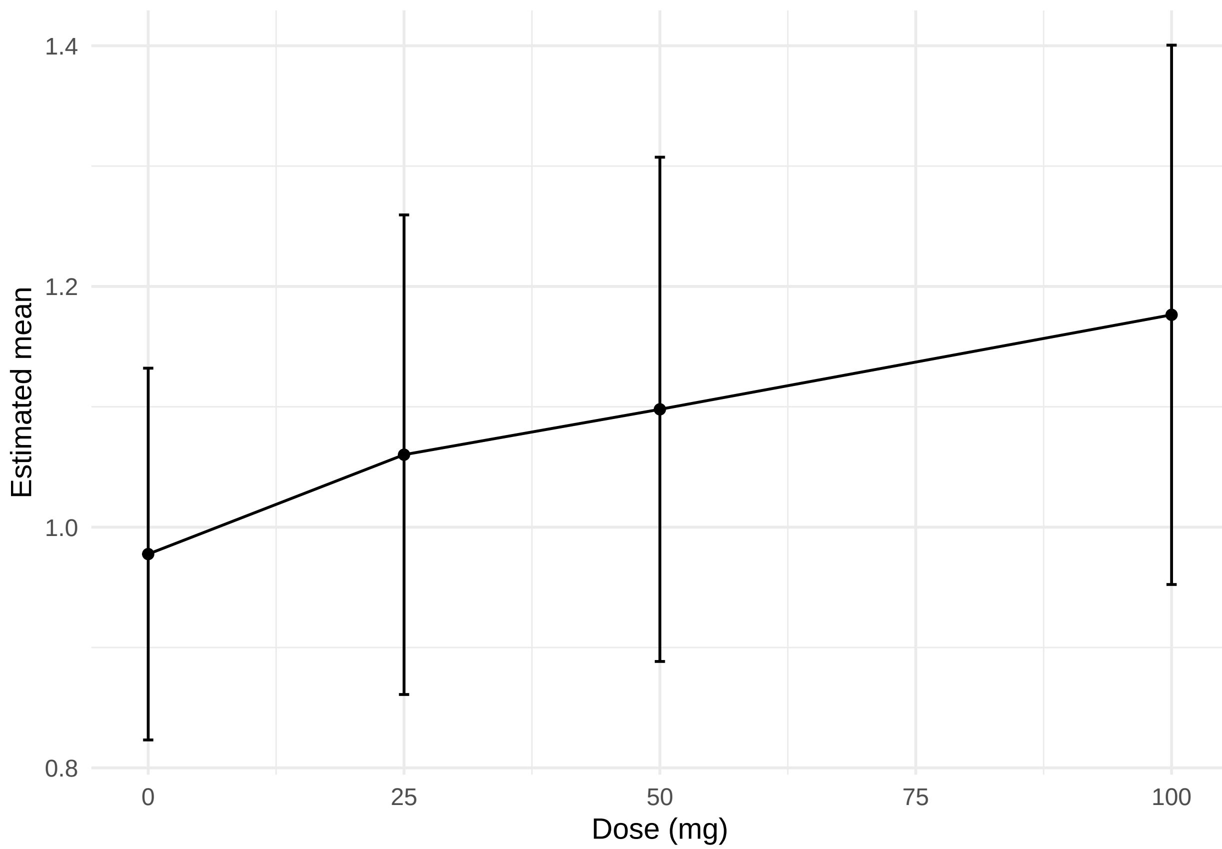
trtcd estimate std.error df conf.low conf.high statistic p.value trt
1 0 0.978 0.0772 61.0 0.823 1.13 12.7 1.02e-18 0 mg
2 25 1.06 0.100 78.1 0.861 1.26 10.6 9.15e-17 25 mg 3 50 1.10 0.105 82.2 0.888 1.31 10.4 1.06e-16 50 mg 4 100 1.18 0.113 84.8 0.952 1.40 10.4 6.93e-17 100 mg # i 1 more variable: mean_txt

Differences vs 0 mg

A tibble: 3 x 11

term contrast null.value estimate std.error df conf.low conf.high 1 trtcd 25 mg vs 0 mg 0 0.0826 0.121 82.7 -0.208 0.373 2 trtcd 50 mg vs 0 mg 0 0.120 0.123 79.6 -0.176 0.416 3 trtcd 100 mg vs 0 mg 0 0.199 0.129 79.9 -0.111 0.508 # i 3 more variables: statistic , adj.p.value , estimate_txt

Plot



mrrvsi

Means by dose

A tibble: 4 x 10

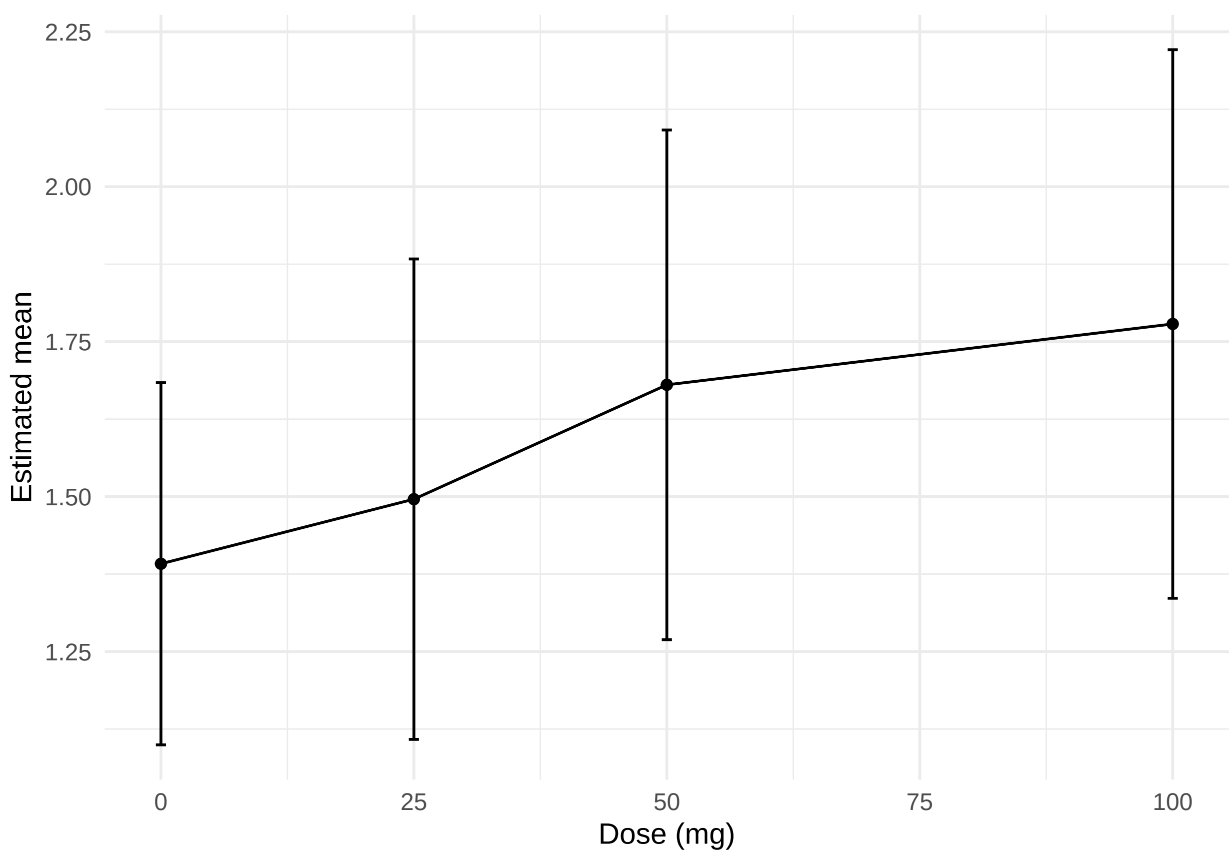
trtcd estimate std.error df conf.low conf.high statistic p.value trt
1 0 1.39 0.146 66.3 1.10 1.68 9.51 5.27e-14 0 mg
2 25 1.50 0.195 77.6 1.11 1.88 7.68 3.95e-11 25 mg 3 50 1.68 0.207 81.0 1.27 2.09 8.13 4.15e-12 50 mg 4 100 1.78 0.223 83.0 1.34 2.22 7.99 6.70e-12 100 mg # i 1 more variable: mean_txt

Differences vs 0 mg

A tibble: 3 x 11

term contrast null.value estimate std.error df conf.low conf.high 1 trtcd 25 mg vs 0 mg 0 0.104 0.249 87.6 -0.494 0.702 2 trtcd 50 mg vs 0 mg 0 0.289 0.256 85.9 -0.326 0.903 3 trtcd 100 mg vs 0 mg 0 0.387 0.267 86.3 -0.255 1.03 # i 3 more variables: statistic , adj.p.value , estimate_txt

Plot



mrrvt

Means by dose

A tibble: 4 x 10

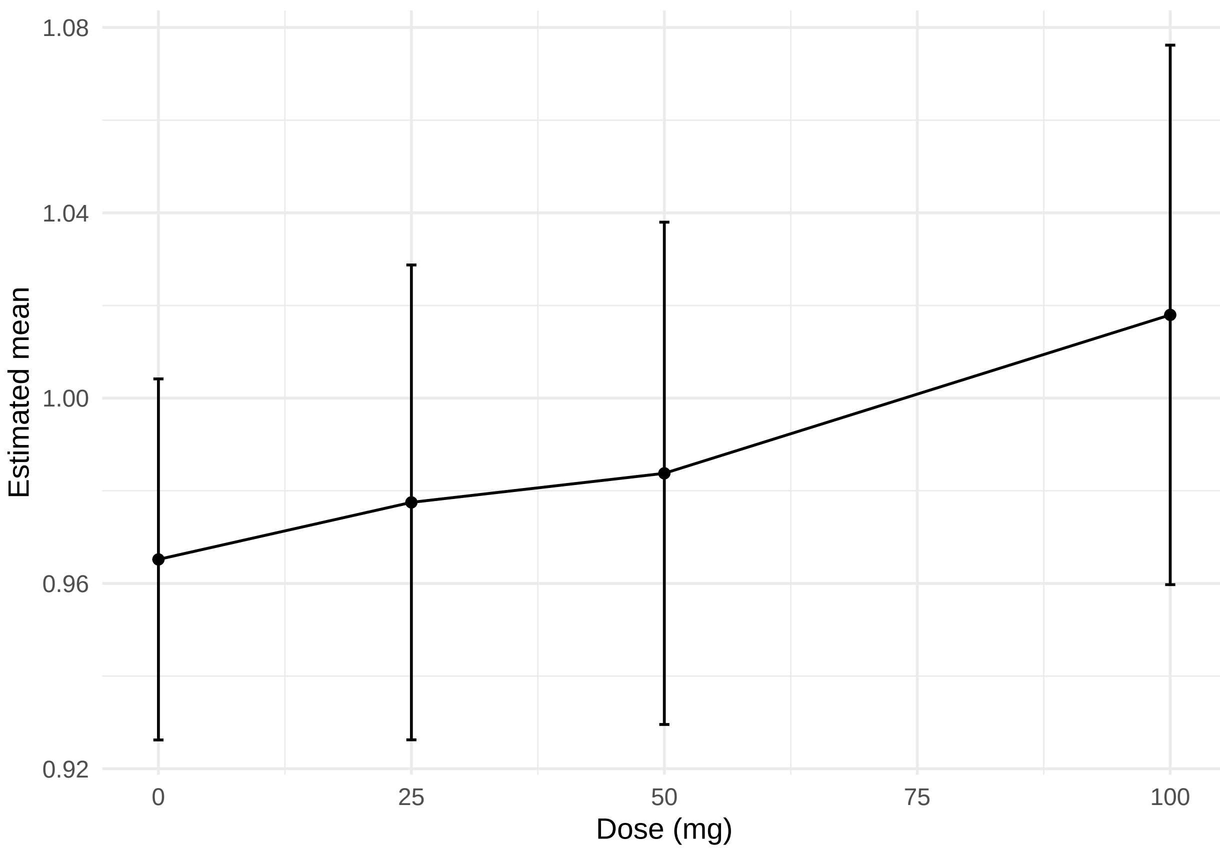
trtcd estimate std.error df conf.low conf.high statistic p.value trt
1 0 0.965 0.0195 64.8 0.926 1.00 49.5 3.64e-53 0 mg
2 25 0.977 0.0257 78.2 0.926 1.03 38.0 4.00e-52 25 mg 3 50 0.984 0.0273 81.9 0.930 1.04 36.1 4.81e-52 50 mg
4 100 1.02 0.0293 84.1 0.960 1.08 34.8 1.14e-51 100 mg # i 1 more variable: mean_txt

Differences vs 0 mg

A tibble: 3 x 11

term contrast null.value estimate std.error df conf.low conf.high 1 trtcd 25 mg vs 0 mg 0 0.0123 0.0323 86.1 -0.0653 0.0899 2 trtcd 50 mg vs 0 mg 0 0.0186 0.0331 83.8 -0.0609 0.0981 3 trtcd 100 mg vs 0 mg 0 0.0528 0.0346 84.2 -0.0303 0.136 # i 3 more variables: statistic , adj.p.value , estimate_txt

Plot



mrdte

Means by dose

A tibble: 4 x 10

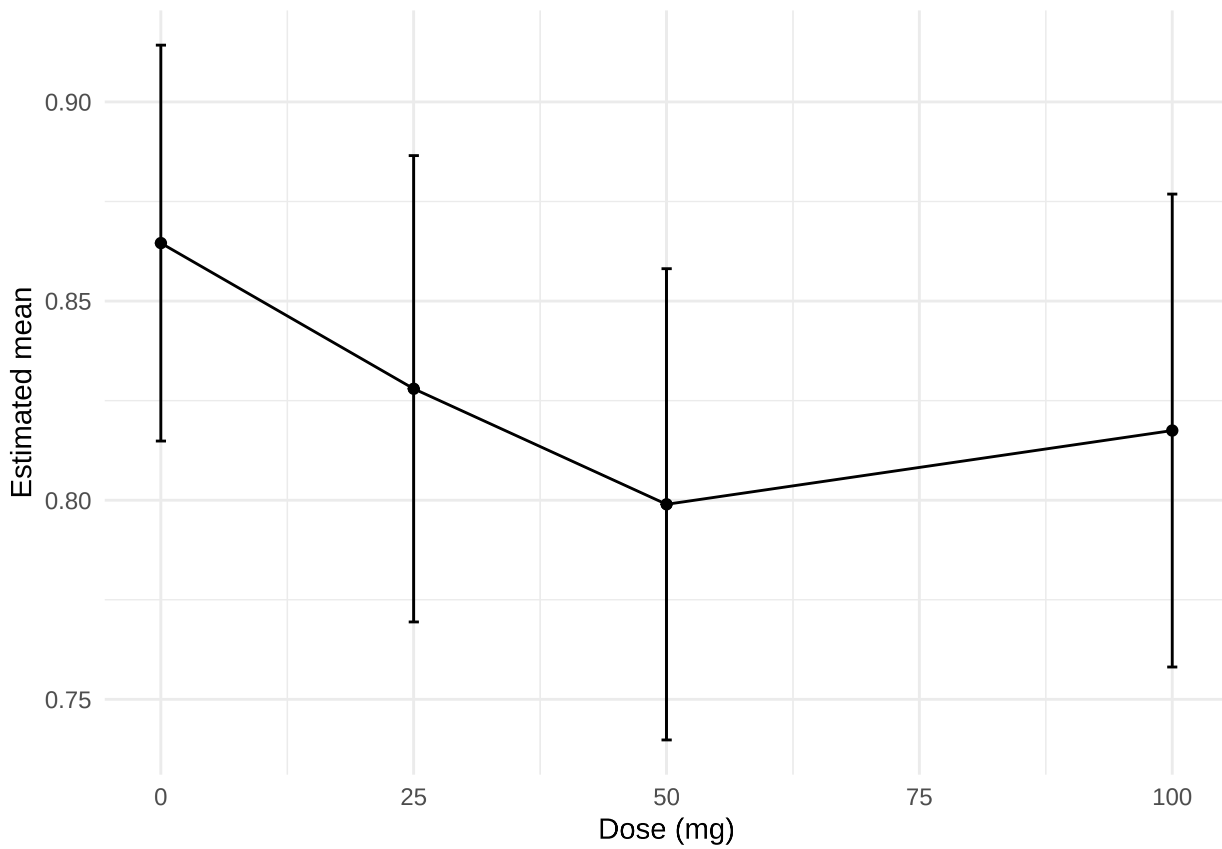
trtcd estimate std.error df conf.low conf.high statistic p.value trt
1 0 0.865 0.0249 71.4 0.815 0.914 34.7 2.08e-46 0 mg
2 25 0.828 0.0295 98.3 0.769 0.887 28.1 1.01e-48 25 mg 3 50 0.799 0.0298 104. 0.740 0.858 26.8 1.69e-48 50 mg 4 100 0.817 0.0299 103. 0.758 0.877 27.3 4.65e-49 100 mg # i 1 more variable: mean_txt

Differences vs 0 mg

A tibble: 3 x 11

term contrast null.value estimate std.error df conf.low conf.high 1 trtcd 25 mg vs 0 mg 0 -0.0366 0.0307 101. -0.110 0.0369 2 trtcd 50 mg vs 0 mg 0 -0.0656 0.0294 93.1 -0.136 0.00492 3 trtcd 100 mg vs 0 mg 0 -0.0471 0.0299 94.4 -0.119 0.0246 # i 3 more variables: statistic , adj.p.value , estimate_txt

Plot



mrdtrv

Means by dose

A tibble: 4 x 10

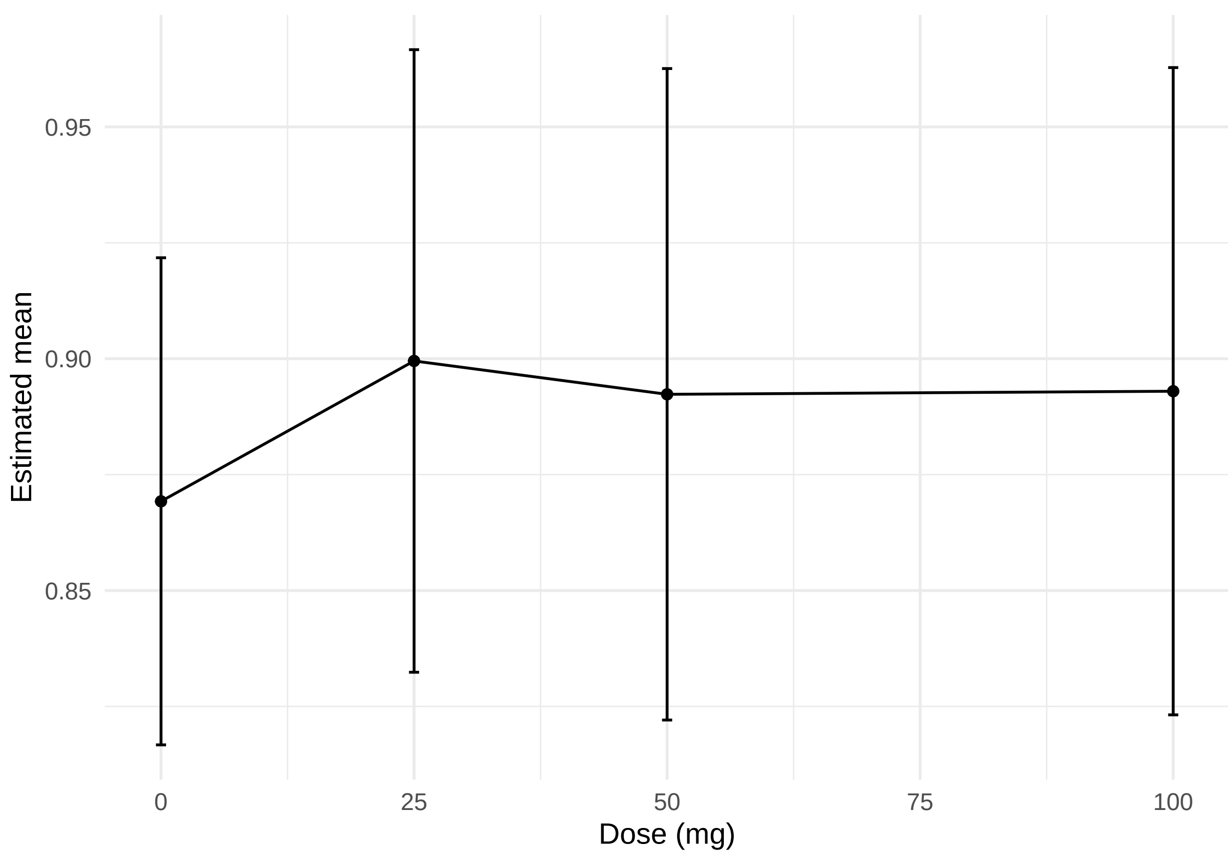
trtcd estimate std.error df conf.low conf.high statistic p.value trt
1 0 0.869 0.0265 99.1 0.817 0.922 32.8 4.68e-55 0 mg
2 25 0.900 0.0339 110. 0.832 0.967 26.5 9.09e-50 25 mg 3 50 0.892 0.0355 120. 0.822 0.963 25.1 1.10e-49 50 mg 4 100 0.893 0.0352 115. 0.823 0.963 25.3 5.68e-49 100 mg # i 1 more variable: mean_txt

Differences vs 0 mg

A tibble: 3 x 11

term contrast null.value estimate std.error df conf.low conf.high 1 trtcd 25 mg vs 0 mg 0 0.0303 0.0436 122. -0.0738 0.134 2 trtcd 50 mg vs 0 mg 0 0.0231 0.0435 112. -0.0810 0.127 3 trtcd 100 mg vs 0 mg 0 0.0237 0.0439 116. -0.0811 0.129 # i 3 more variables: statistic , adj.p.value , estimate_txt

Plot



mrdts

Means by dose

A tibble: 4 x 10

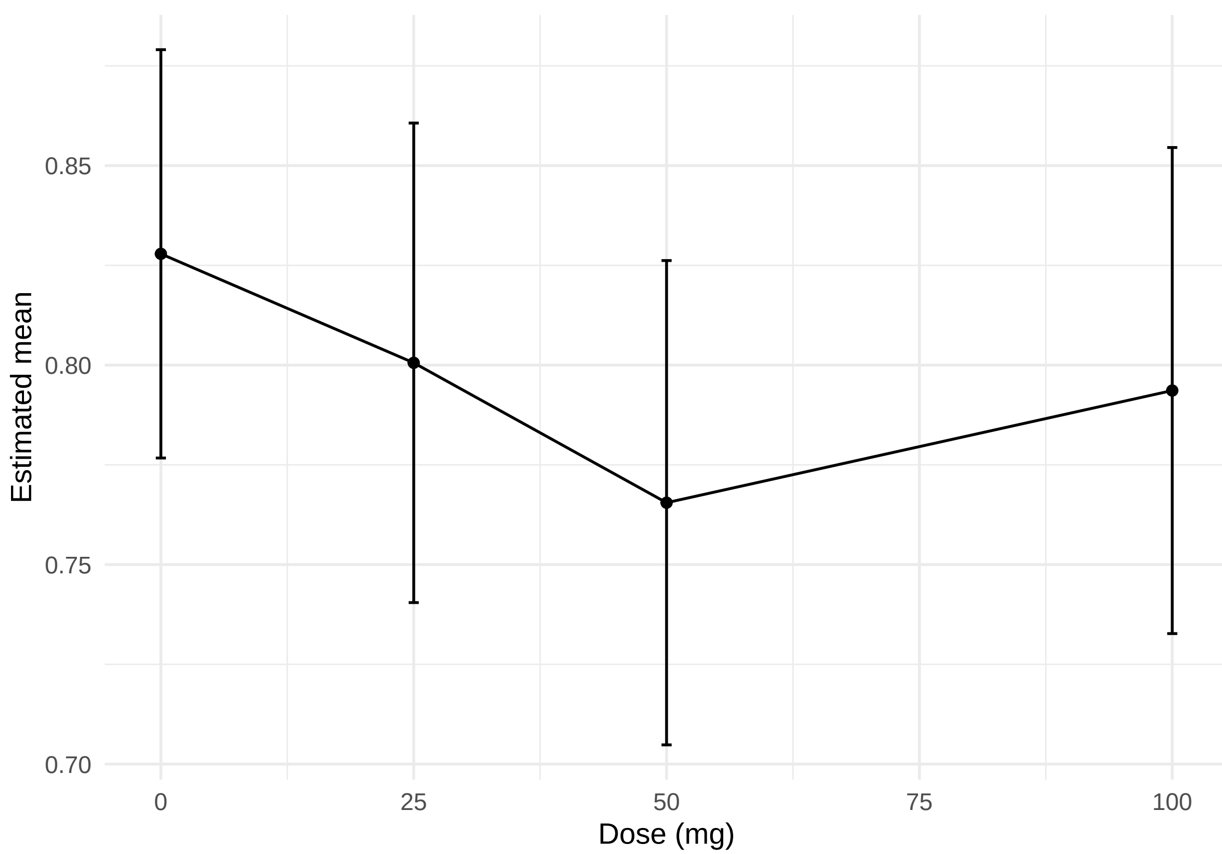
trtcd estimate std.error df conf.low conf.high statistic p.value trt
1 0 0.828 0.0257 70.6 0.777 0.879 32.3 5.63e-44 0 mg
2 25 0.801 0.0303 97.4 0.740 0.861 26.4 3.06e-46 25 mg 3 50 0.766 0.0306 103. 0.705 0.826 25.0 1.41e-45 50 mg 4 100 0.794 0.0307 102. 0.733 0.855 25.8 1.17e-46 100 mg # i 1 more variable: mean_txt

Differences vs 0 mg

A tibble: 3 x 11

term contrast null.value estimate std.error df conf.low conf.high 1 trtcd 25 mg vs 0 mg 0 -0.0273 0.0312 101. -0.102 0.0475 2 trtcd 50 mg vs 0 mg 0 -0.0624 0.0299 92.7 -0.134 0.00933 3 trtcd 100 mg vs 0 mg 0 -0.0343 0.0304 94.0 -0.107 0.0386 # i 3 more variables: statistic , adj.p.value , estimate_txt

Plot



mrdtv

Means by dose

A tibble: 4 x 10

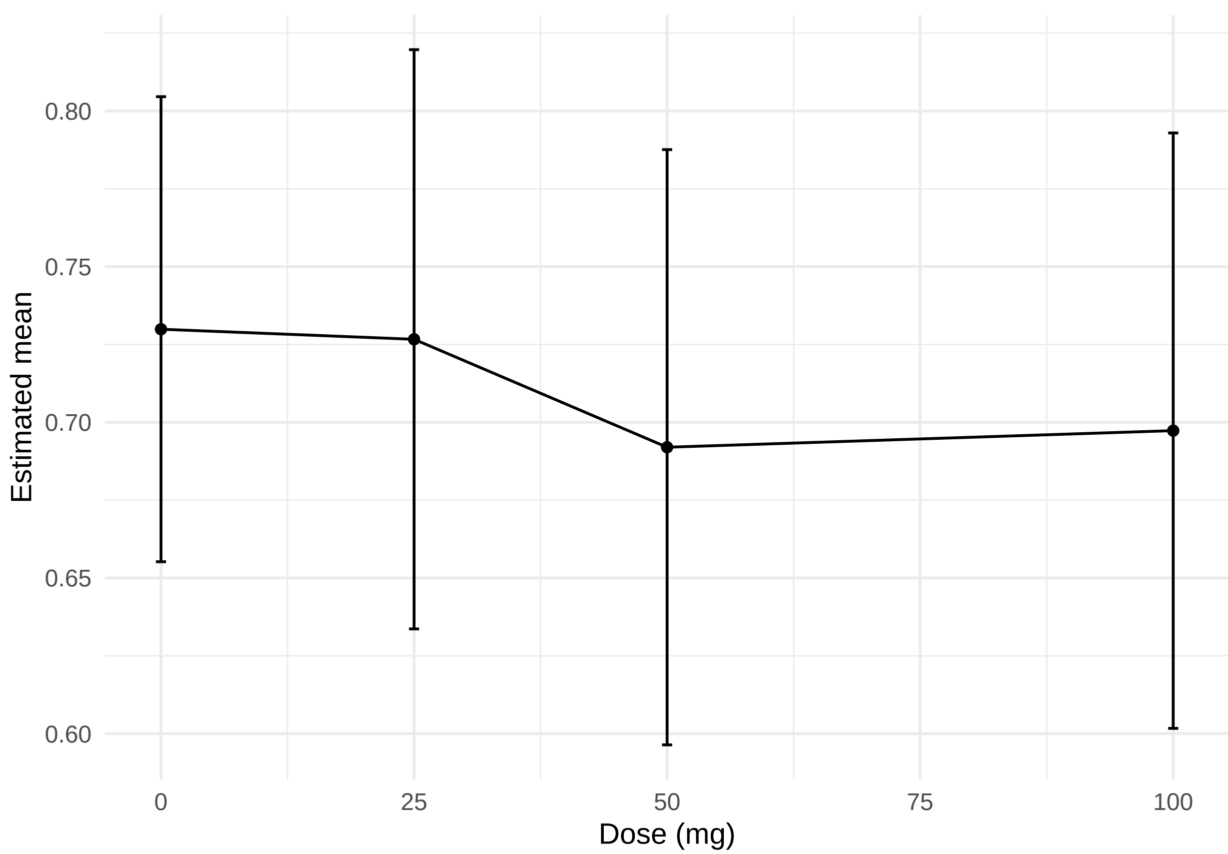
trtcd estimate std.error df conf.low conf.high statistic p.value trt
1 0 0.730 0.0376 89.0 0.655 0.805 19.4 9.47e-34 0 mg
2 25 0.727 0.0469 110. 0.634 0.820 15.5 2.42e-29 25 mg 3 50 0.692 0.0483 118. 0.596 0.788 14.3 1.33e-27 50 mg 4 100 0.697 0.0483 115. 0.602 0.793 14.4 1.23e-27 100 mg # i 1 more variable: mean_txt

Differences vs 0 mg

A tibble: 3 x 11

term contrast null.value estimate std.error df conf.low conf.high 1 trtcd 25 mg vs 0 mg 0 -0.00324 0.0562 113. -0.138 0.131 2 trtcd 50 mg vs 0 mg 0 -0.0379 0.0550 102. -0.170 0.0938 3 trtcd 100 mg vs 0 mg 0 -0.0326 0.0557 105. -0.166 0.101 # i 3 more variables: statistic , adj.p.value , estimate_txt

Plot



mrt1ca

Means by dose

A tibble: 4 x 10

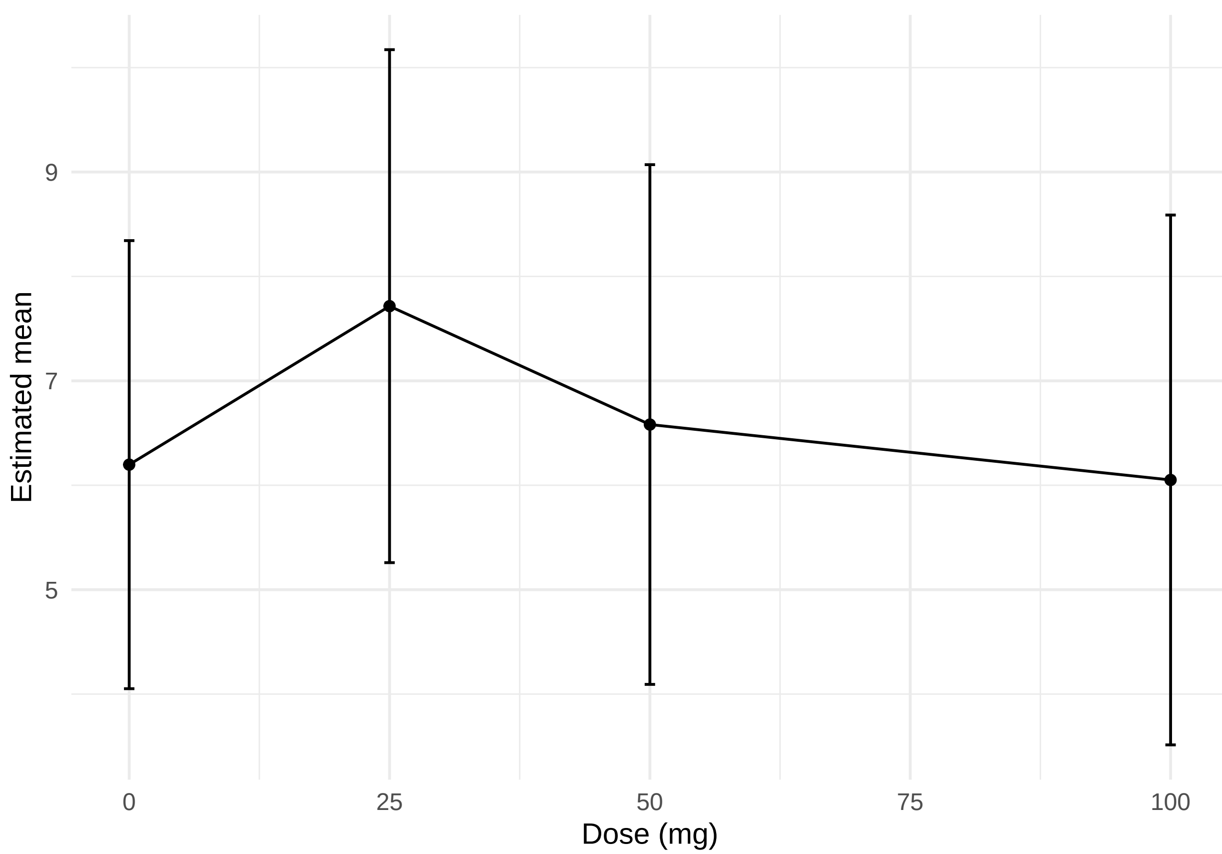
trtcd estimate std.error df conf.low conf.high statistic p.value trt
1 0 6.20 1.07 65.3 4.05 8.34 5.77 0.000000239 0 mg 2 25 7.72 1.24 99.5 5.26 10.2 6.23 0.0000000112 25 mg 3 50 6.58 1.25 103. 4.09 9.07 5.24 0.000000839 50 mg 4 100 6.05 1.28 108. 3.51 8.59 4.73 0.00000692 100 ~ # i 1 more variable: mean_txt

Differences vs 0 mg

A tibble: 3 x 11

term contrast null.value estimate std.error df conf.low conf.high 1 trtcd 25 mg vs 0 mg 0 1.52 1.06 135. -1.02 4.06 2 trtcd 50 mg vs 0 mg 0 0.384 1.08 135. -2.19 2.96 3 trtcd 100 mg vs 0 mg 0 -0.147 1.11 135. -2.79 2.50 # i 3 more variables: statistic , adj.p.value , estimate_txt

Plot



mrt2ps

Means by dose

A tibble: 4 x 10

trtcd estimate std.error df conf.low conf.high statistic p.value trt
1 0 20.6 2.96 76.1 14.7 26.5 6.95 1.09e-9 0 mg 2 25 18.0 3.59 120. 10.9 25.1 5.01 1.85e-6 25 mg 3 50 15.0 3.65 125. 7.74 22.2 4.10 7.47e-5 50 mg 4 100 19.0 3.75 130. 11.6 26.4 5.07 1.35e-6 100 ~ # i 1 more variable: mean_txt

Differences vs 0 mg

A tibble: 3 x 11

term contrast null.value estimate std.error df conf.low conf.high 1 trtcd 25 mg vs 0 mg 0 -2.58 3.54 141. -11.0 5.86 2 trtcd 50 mg vs 0 mg 0 -5.61 3.59 141. -14.2 2.95 3 trtcd 100 mg vs 0 mg 0 -1.58 3.68 141. -10.4 7.20 # i 3 more variables: statistic , adj.p.value , estimate_txt

Plot